ENGINEERING SCIENCE

Revised
July 2022

Pupil booklet

Mechanisms and structures:

Pneumatics and control

Topic 1: Explain pneumatics and its place in the World.

Topic 2: Introduce components, circuits and control.

Topic 3: Pneumatic calculations Force = Pressure x Area.

Topic 4: Extension: Computer control – flowcode. (Book 2.)



Name: _____

Class: _____

Teacher: _____

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Unit outline

The general aim of this Unit is to develop an understanding of mechanisms and structures. Learners will explore a range of mechanical and pneumatic systems and design, simulate, construct, test and evaluate mechanical or pneumatic solutions to solve problems.

Learners who complete this unit will be able to:

1 Investigate a range of mechanical and pneumatic systems

2 Develop mechanical or pneumatic solutions to solve problems

Outcomes and assessment standards

Outcome 1

The learner will:

1 Investigate a range of pneumatic systems by:

1.2 Describing or producing diagrams of a range of pneumatic systems.

Outcome 2

The learner will:

2 Develop mechanical or pneumatic solutions to solve problems by:

2.1 Identifying key aspects of a problem.

2.2 Applying knowledge and understanding of structures, pneumatics and/or mechanical drive systems.

2.3 Designing, and simulating or building, mechanical or pneumatic systems.

2.4 Testing and evaluating solutions against a specification.

Success criteria

Use traffic lighting to record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will sign off this page after speaking to you about your self-evaluation:

Red: I need to work on this topic further to understand it.

Amber: I need a bit of help to fully understand this topic.

Green: I am confident in my understanding of this topic.

Success criteria

- ☐ I can identify 3 advantages and 3 disadvantages of using compressed air.
- ☐ I know what safety precautions to take when using compressed air.
- ☐ I can recognise and draw symbols used in pneumatic circuits.
- ☐ I can describe the function of several components used in pneumatic systems
- ☐ I can explain the terms in-stroke and out-stroke in relation to Pneumatic Control.
- ☐ I know how to create AND & OR control within a pneumatic circuit.
- ☐ I can create a circuit that works in a set sequence.
- ☐ I can create circuits that are manual and automatically controlled.
- ☐ I can create a time delays in pneumatics.
- ☐ I can do calculations using force, pressure and area.

What is a pneumatic system?

Pneumatics is something that you probably know very little about yet come across everyday without ever realising it. Some examples are shown below.





Pneumatics are used a lot in industry and you would expect to see pneumatics systems in factories, production lines and processing plants. It can be used to do many different jobs such as moving, holding or shaping objects. Even dentist equipment is pneumatic operated.

Every one of these pneumatic systems only makes use of compressed air. In the classroom, you are supplied with compressed air through a *manifold.* The manifold lets you connect lots of components to the compressed air.

Safety Rules:

Safety rules help keep us safe. When we are using pneumatics, we must follow these rules.

1. Never blow compressed air at anyone, not even yourself
2. Never let compressed air come into contact with your skin, as this can be very dangerous.
3. Always wear safety goggles when you are connecting and operating circuits.
4. Check that all airlines are connected before turning on the main air supply. If you hear air then a circuit has a leak and it must be located.
5. Never disconnect a circuit when the system is pressurised. Always turn off the main air supply and release the pressure before disconnecting a pipe circuit.
6. Disconnect a pipe circuit from the *manifold* first and not the individual component.
7. Keep your hands away from moving parts.
8. Avoid having airlines trailing across the floor or where someone could trip or become entangled.

Pneumatic colour code:

You must follow the following colour codes when building a system. Using the correct colour code will assist you when you need to explain the system.

Red	Operating system pressure.
Blue	Exhaust- flow without restriction.
Green	Intake from reservoir from pump.
Yellow	Controlled flow.

Use this space below to SIGN and DATE the reading and understanding of following the safety rules. Failure to follow will result in you being suspended from using the equipment.

Advantages of pneumatic systems

There are usually lots of different ways to carry out a task, so it is important to understand some of the reasons for choosing pneumatic systems.

Clean

Pneumatic systems are clean because they use compressed air. We know already that this is just the air we breathe forced into small spaces. If a pneumatic system develops a leak, it will be air that escapes and not oil. This air will not drip or cause a mess and this makes pneumatics suitable for food production lines.

Safe

Pneumatic systems are very safe compared to other systems. We cannot use electronics for paint spraying because many electronic components produce sparks and this could cause the paint to catch fire. It is important, however, that we look after and maintain the different components. It is also important that we follow the correct safety rules.

Reliable

Pneumatic systems are very reliable and can keep working for a long time. Many companies invest in pneumatics because they know they will not have a lot of breakdowns and that the equipment will last for a long time.

Economical

If we compare pneumatic systems to other systems, we find that they are cheaper to run. This is because the components last for a long time and because we are using compressed air. Many factories already have compressed air for other reasons.

Flexible

Once you have bought the basic components, you can set them up to carry out different tasks. Pneumatic systems are easy to install and they do not need to be insulated or protected like electronic systems.

Disadvantages of pneumatic systems

Small forces

Pneumatic components produce relatively small forces. Whenever large forces are required, hydraulic systems are used. Hydraulic systems contain liquid which, unlike air, cannot be compressed. This means they can create large forces, and that the force and speed of movement are more controllable.

Not energy efficient

Compared with an electrical system, a pneumatic system is not very energy-efficient. A lot of energy is converted in the production of compressed air at the input end of the system, compared with the amount of energy finally converted into movement at the output end of the system.

Lack of intermediate control

Pneumatics is very good at providing linear motion, especially when a pneumatic piston is pushed fully out or fully in. However, it is not very good at intermediate positioning, i.e., it is very difficult for the piston to be part of the way out or part of the way in.

Consequences of failure

If the air supply was to fail in the middle of lifting a heavy load, this could have disastrous consequences.

Components - cylinders

Pneumatic equipment can be split up into two basic categories of cylinders and valves.

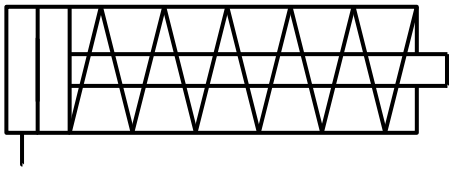
Cylinders are the 'muscles' of pneumatic systems as they are used to move, hold and lift objects. They can even be used to operate other pneumatic components. Cylinders are operated by compressed air and they covert the stored energy in the compressed air into *linear* motion.

Linear motion is motion in a straight line: an apple falling from a tree or a sliding door closing are examples of linear motion. We can represent linear motion by arrows like the ones below.

There are two types of cylinder that we will be using: single-acting cylinders and double-acting cylinders.

Single-acting cylinder, spring return

A single-acting cylinder requires only one air supply. If we supply compressed air to a single-acting cylinder, the air pushes against the piston inside the cylinder and causes it to *outstroke*. When the piston has fully *outstroked* it is said to be *positive*. If we stop the supply of air then the spring (illustrated by zig zags) inside the cylinder causes the piston to *instroke* to its starting position and the piston is said to be *negative*. As this happens, the air inside the cylinder is pushed back out.



Double-acting cylinder

A double-acting cylinder has **no** spring inside to return it to its original position. It needs two air supplies, one to *outstroke* the piston and the other to *instroke* the piston.

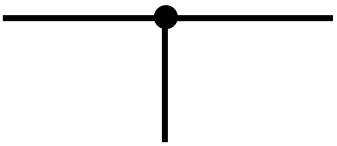
To outstroke a double-acting cylinder we need compressed air to push against the piston inside the cylinder. As this happens, any air on the other side of the piston is forced out. This causes the double-acting cylinder to *outstroke*. When the piston has fully outstroked it is said to be *positive*.

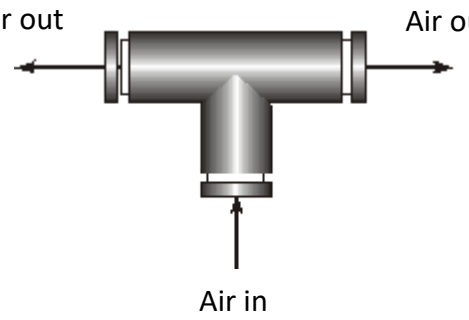


To instroke a double-acting cylinder we need to reverse this action. We supply the compressed air to the other side of the piston. As the air pushes the piston back to its original position, any air on the other side is again forced out. This causes the piston to *instroke* and it is said to be *negative*.

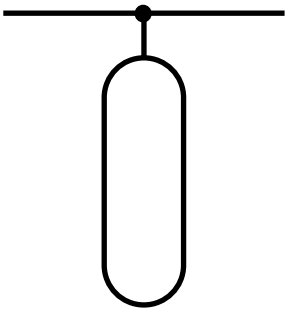
T-Piece

A T-Piece T-connector is a very simple component that lets us split or divide airflow. It can be very useful if you want two cylinders to operate at the same time. On circuit diagrams, the T-Piece is identified by a dot as shown below;





Time delays



Sometimes in a system we want a pause or delay before something else happens. To create a delay we need to use two components; a uni-directional restrictor and a reservoir.

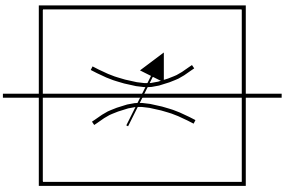
A reservoir is simply an empty container, just like an empty bottle. The bigger the reservoir, the longer it takes to fill up with air. To make the delay longer we use a unidirectional restrictor in front of the reservoir. This slows down the air so that the reservoir takes even longer to fill. The length of time it takes to fill creates the delay. We can change the length of a delay by changing the size of the reservoir or adjusting the restrictor.

Time delays can be very useful in clamping operations when objects need to be held in place by a cylinder for a specific amount of time to glue or set.

Flow control valves (SPEED CONTROL)

Some of the circuits you will build will result in the pistons moving very quickly. This can be dangerous or it may prevent a circuit working correctly. To slow down the speed of the pistons we use flow control valves.

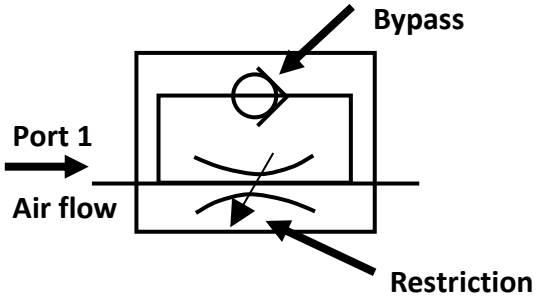
There are TWO types of flow control valve available to us.



The first type is called a ***restrictor*** (or sometimes a throttle valve). This valve works by reducing the amount of space that the air can flow through. We can adjust the airflow by turning the small screw on top of the valve. The symbol for a restrictor is shown opposite. This slows down the air going through it in BOTH directions.

The problem with this type of restrictor is that it always slows down the speed of the piston in both directions. In many cases, we would only want EITHER the outstroke or the instroke to be slowed down. Also, if we study the piston movement very carefully, we sometimes find that it is quite jerky – not smooth as we would want it to be. This is caused by friction and why most flow control valves are placed in the exhaust line of a system.

Uni-directional restrictor



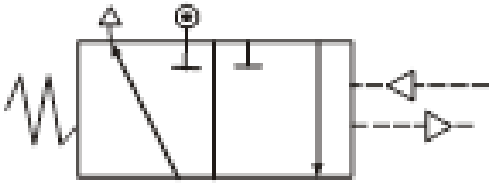
To solve these problems, we can use a component called a *uni-directional restrictor*. As its name suggests, it only slows down the air in one direction. The symbol is shown opposite. When air flows into port 1 of the restrictor, some of the air takes the bypass route. A small ball is blown against a valve and blocks this path. The air is then forced to go through the restriction and this slows down the airflow.

In pneumatics, uni-directional restrictors are much more useful to us. However, we must always be careful to insert them in the circuit the correct way around. It can also be called a throttle valve.

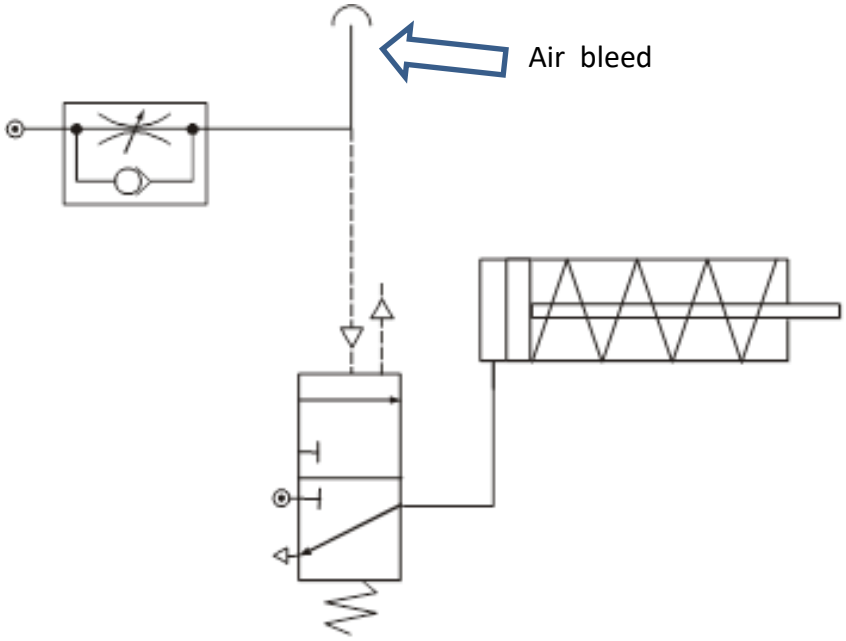
Air bleed

Sometimes with pneumatics we find that the actuators on valves can get in the way of the circuit. Also, some actuators need a big force to make them work and this is not always possible. There are different ways to overcome these problems and one of the most common is to use an *air bleed*.

An air bleed is simply an open pipe that allows the air in the circuit to escape. This air must be at a low pressure, otherwise the pipe would 'wave' about and be dangerous. Air bleed circuits rely on a component called a *diaphragm valve*. This valve is capable of detecting small changes in air pressure. The valve works in the same way as other 3/2 valves; it is only the actuator that is new to us. The symbol is shown below.



The diaphragm (shown above) is a piece of rubber stretched inside the valve. When air flows into the top of the valve, the rubber expands much in the same way as when a balloon is blown up. When the diaphragm expands, it presses down inside the valve and changes its state.



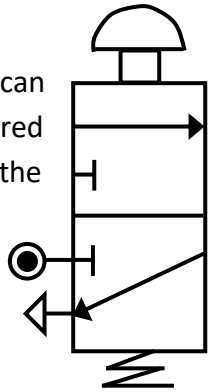
Example:

The signal to the diaphragm comes from an air bleed. When the air bleed is blocked, air is diverted back towards the diaphragm. This actuates the 3/2 valve and the cylinder outstrokes. Notice that the airflow to the air bleed passes through a restrictor. This slows down the air before it is allowed to escape.

Valves

Valves control the flow of compressed air to a cylinder. They can be used to turn the air on or off, change the direction in which the air is flowing or even slow down the airflow. The most common type of valve is the 3/2 valve.

3/2 valve



A 3/2 valve gets its name because it has three ports and two states. A **port** is where we can connect a pipe and a state is simply a position that the valve can be in. The ports are numbered to help us make the right connections. The numbers will be stamped onto the casing of the valve. The full name is a **Push Button Actuated, 3/2 Value, Spring Return.**

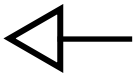
Port identification symbols

Port 1 – mains air - 

This port is connected to main air. Remember that our main air is supplied through a manifold. Main air is identified by this symbol:

Port 2 – output connection -

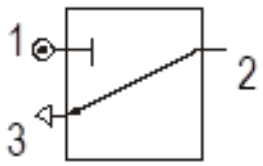
This port lets us make connections to other components. Remember, the purpose of valves is to control the flow of air to other components, usually cylinders.

Port 3 – exhaust air - 

This port allows air trapped in the circuit to escape or exhaust. Remember, for our cylinders to instroke and outstroke, they need the air on the other side of the piston to escape.

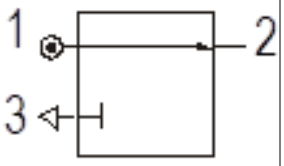
A **state** is simply a position that the valve can be in, of which there is two.

State 1 – off/normal rest position



In this state, the main air supply through the valve is blocked and so air is unable to reach other components, such as cylinders. However, any air within the cylinder is able to exhaust through the valve and this will allow the cylinder to return to its original position.

State 2 – on/actuated

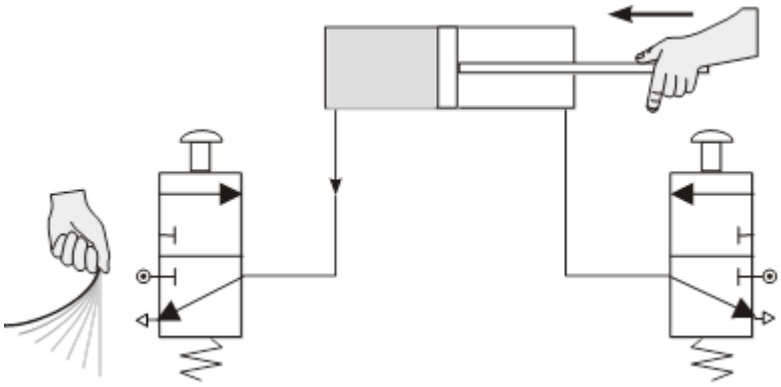


In this state, the main air supply is able to flow freely through the valve and supply components, such as cylinders, with air.

The internal STATE 'shuttles' or 'slides' left and right within the component. One state prevents air from being supplied to other components and the other allows the air to flow freely.

5/2 valve

There are many problems when controlling a double-acting cylinder with two 3/2 valves. You should have discovered from the last circuit that it is easy to push or pull the piston. This is because you do not have a constant supply of air to keep the piston in place. When you actuate the 3/2 valve, it outstrokes the piston. When the 3/2 valve is not actuated, air is free to escape or exhaust back through the valve. This means that any force or effort placed on the piston will make it move easily. Test it yourself to feel how easy it is to move the piston.



A further disadvantage is that the 3/2 valve needs to be actuated until the double-acting cylinder has fully outstroked or instroked. Releasing the valve before the stroke is complete will mean the piston will stop short of its final position.

We have greater control over a double-acting cylinder if we control its outstroke and instroke using a 5/2 valve. This valve has five ports and two states of operation. The ports are always numbered in the same way.

Port 1 – main air

This port is connected to main air just like a 3/2 valve.

Port 2 – output connection

This port is usually connected to instroke a double-acting cylinder.

Port 3 – exhaust

This port allows air trapped in the double-acting cylinder to escape. Remember, for the cylinder to instroke and outstroke, air on the other side of the piston must be allowed to escape. If this did not happen, the piston would not move.

Port 4 – output connection

This port is usually connected to outstroke a double-acting cylinder.

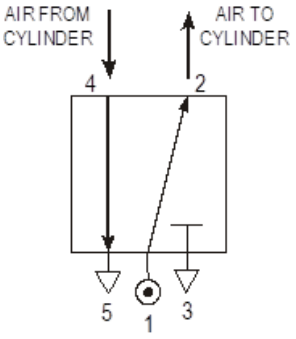
Port 5 – exhaust

Again, this port lets the air on the other side of the piston escape.

Value operation

A 5/2 valve has **five** connection ports and **two** states of operation. One state supplies air to outstroke a double-acting cylinder and the other state will cause it to instroke. The 'pilot air' ports on either side of a 5/2 valve are **not** counted. There are also 7/2 valves.

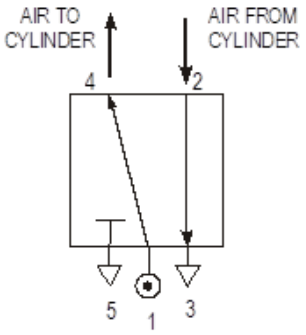
State 1 – Instroke



In this state, the main air flows through the valve from port 1 to port 2. Any air within the cylinder is able to exhaust through the valve from port 4 to port 5. In this state, a 5/2 valve will cause a double-acting cylinder to instroke or hold the piston in the negative position. This means that air is always being supplied to the cylinder.

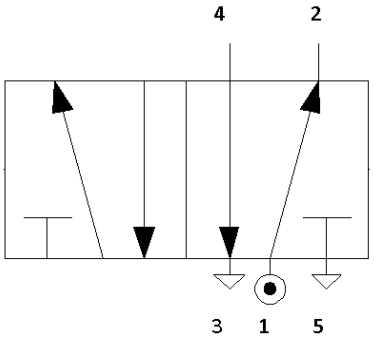
Study the symbol on the right and ensure that you understand how the air flows through the valve ports.

State 2 – Outstroke



In this state, the main air flows through the valve from port 1 to port 4. Any air on the other side of the piston is able to exhaust through the valve from port 2 to port 3. In this state, a 5/2 valve will cause a double-acting cylinder to outstroke and will hold it in the 'positive' position.

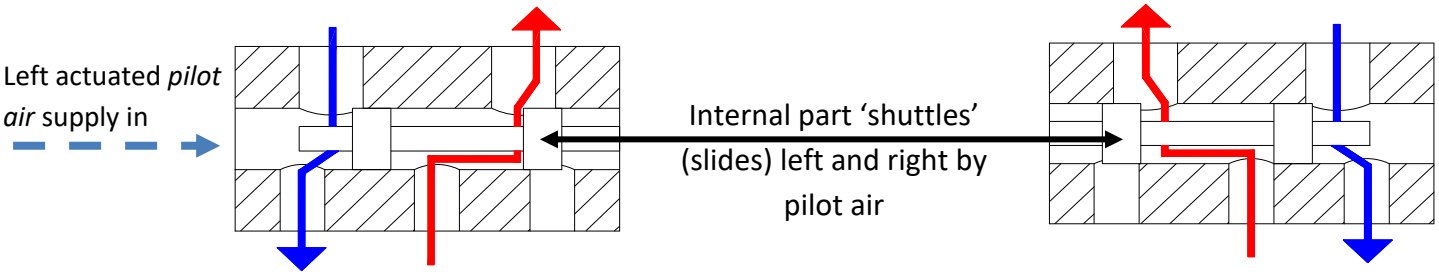
Study the symbol opposite and ensure that you understand how the air flows through the valve.



The complete symbol for a 5/2 valve is shown opposite. The cylinder, main air and exhaust lines are only ever drawn on one half of the symbol, depending on which part is active in the circuit. It is usual, however, to draw the symbol so that it would hold a double-acting cylinder in the negative position.

Inside a 5/2 valve.

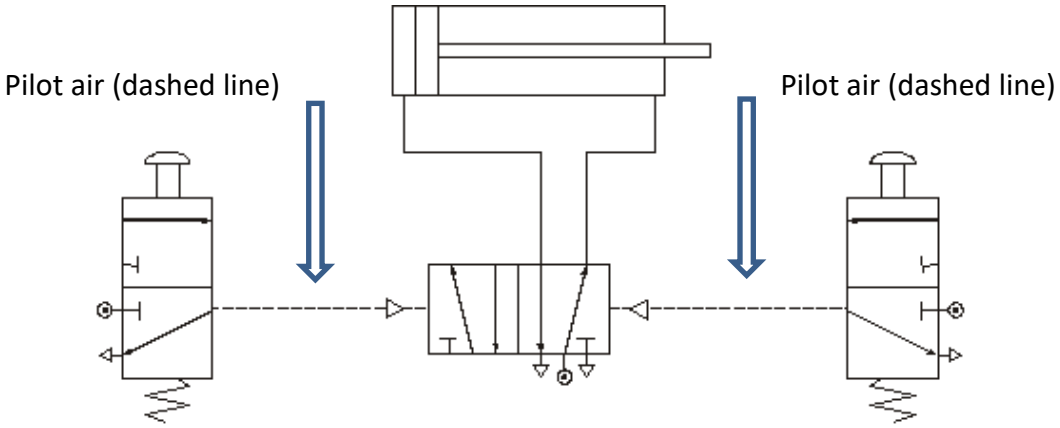
Valve 'shuttled' to the right by left pilot air. *Valve 'shuttled' to the left by right pilot air.*



Exhaust (3) Mains air (1) feeding position (2) *Mains air (1) Exhaust (5)*

Pilot air ----------

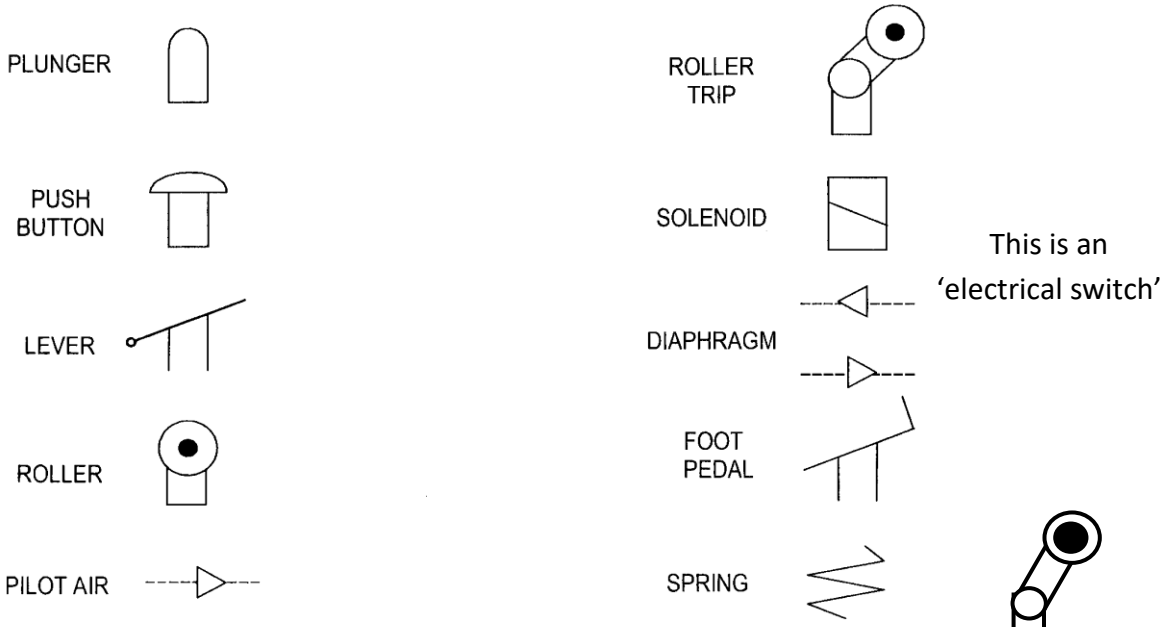
5/2 valves can be operated or actuated in the same way as 3/2 valves. However, the most common way of actuating a 5/2 valve is by pilot air. A pilot air 5/2 valve will change state when a brief air signal acts at either end of the valve. This signal is most often supplied from a 3/2 valve. In the example shown below, the button on valve A only needs to be pressed for a moment in order to change the state of the 5/2 valve. The 5/2 valve supplies the double-acting cylinder with air to make it outstroke.



Notice that the pilot airlines to the 5/2 valve are drawn as broken or dashed lines to distinguish them from the other air lines in the circuit.

Actuators

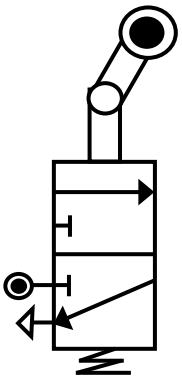
There are a number of different ways we can operate a 3/2 valve. The most common types of actuators are shown below.



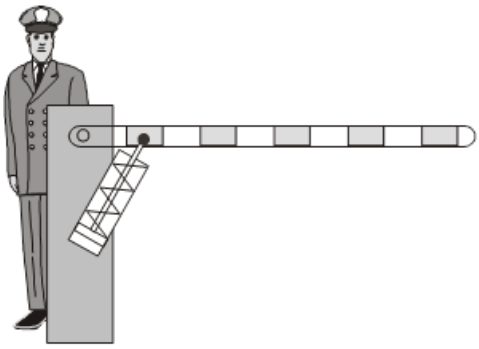
When naming the component, you must state its **FULL** name.

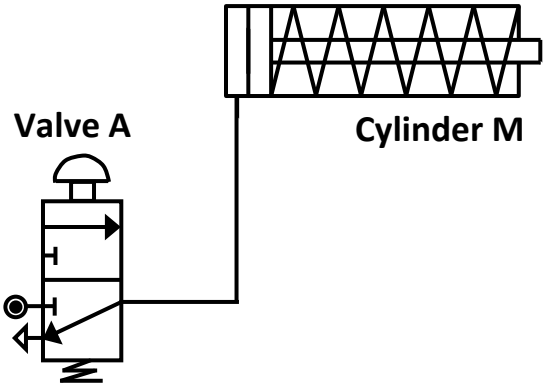
Example

This one would be known as a '**Roller trip actuated, 3/2 valve, spring return**'



1. A city-centre car park has a barrier system to prevent people parking illegally. The car park task attendant checks all the cars entering and leaving the car park. The barrier is raised and lowered by a single-acting cylinder. The attendant pushes a button to operate the system. Build and test the circuit below. A full named component example of how to explain the circuit has been provided below. Further system descriptions of systems can be obviated to just Valve A, Valve B, Cylinder M and so on.



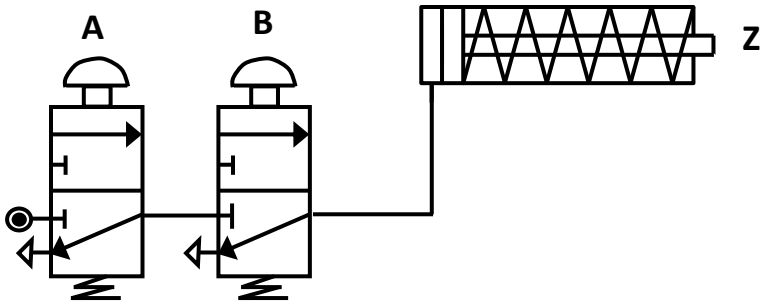


"When the push button 3/2 spring return (valve A) is actuated (pressed down), the valve changes state and supplies main air to the single-acting spring return cylinder (Cylinder M) (1 mark). This causes the piston to outstroke (positive) with force to raise the barrier (1 mark). When the button is released, the valve returns to its original state and stops the air supply which causes the piston to instroke (negative) (1 mark). The system resets and is ready for the process to begin again".

AND control

Although pneumatic circuits are very safe, it is important to take safety precautions. 'AND' control circuits can be used to help prevent accidents by ensuring that guards are in position before machines are switched on. These circuits can also be used to stop a machine being switched on accidentally or to stop operators placing their hands in the machine when it is running.

AND control involves connecting 3/2 valves together in *series*. This means that the output from one valve becomes the input to another. We can summarise how the circuit behaves in a *truth table*.



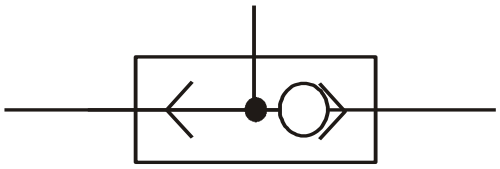
Truth table for **AND** gate

A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1

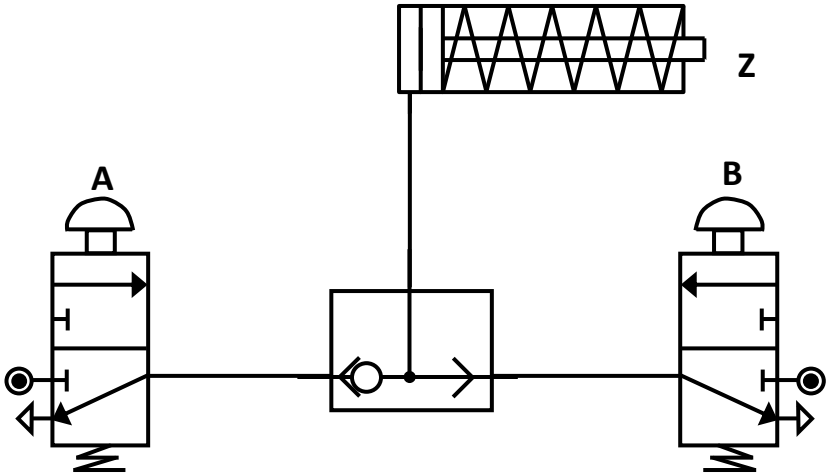
2. Build and test the circuit above. In your jotter, draw the circuit and using the correct terminology, explain exactly how this circuit works.

OR control

Sometimes we need to control a pneumatic circuit from more than one position. This can be done using 'OR' control circuits. These circuits are quite simple but they need another component called a ***'shuttle'*** valve (shown below).



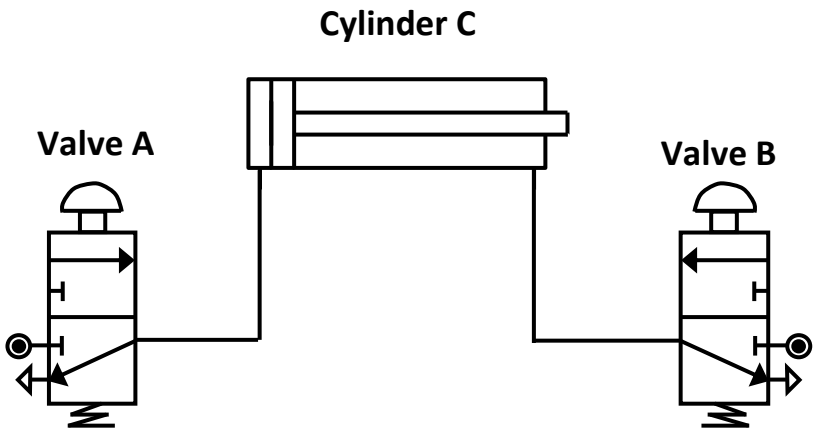
A shuttle valve is used to change the direction of air in a circuit. It has a small ball inside that gets blown from side to side. OR control involves connecting 3/2 valves together in *parallel*. This means that either valve will outstroke the cylinder.



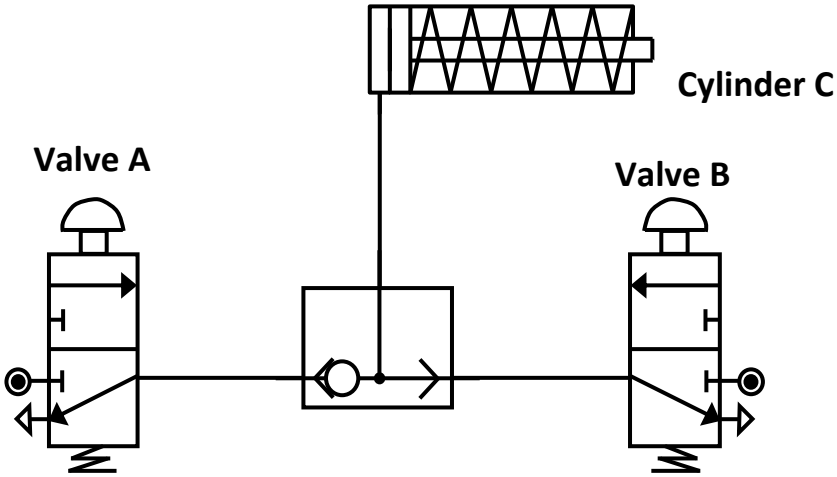
A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

Truth table for OR gate

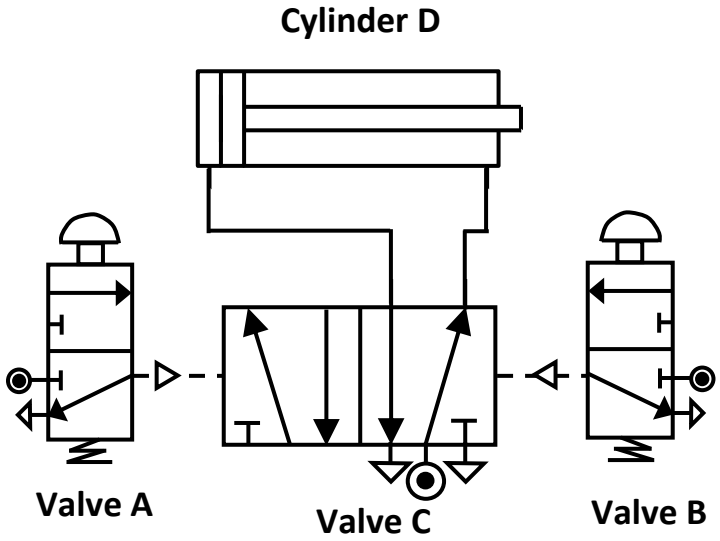
3. Build and test the circuit below. In your jotter, draw the circuit and using the correct terminology, explain exactly how this circuit works.



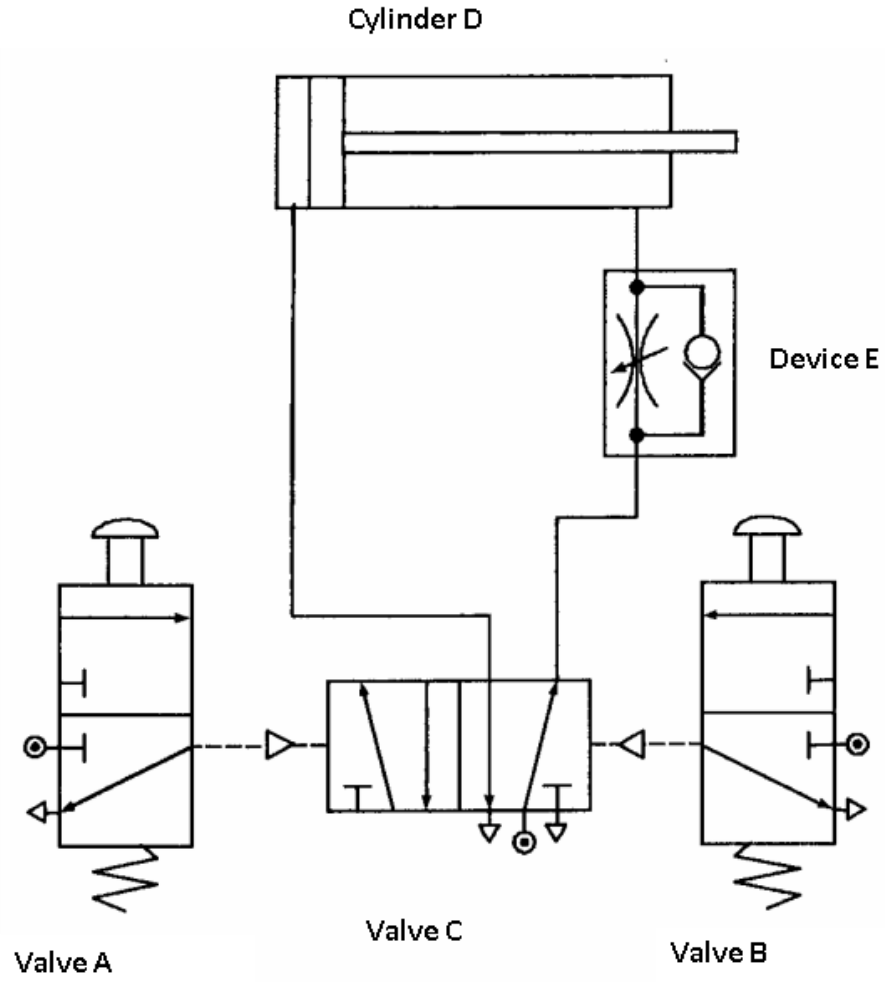
4. Build and test the circuit below. In your jotter, draw the circuit and using the correct terminology, explain exactly how this circuit works.



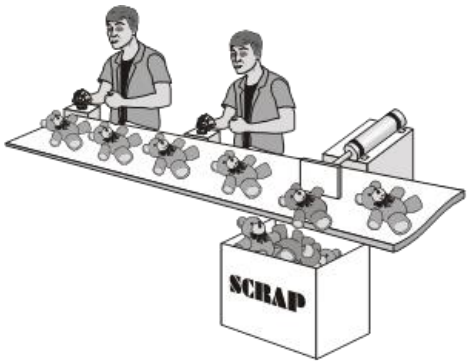
5. Build and test the circuit opposite. In your jotter, draw the circuit and using the correct terminology, explain exactly how this circuit works.



6. Build and test the circuit below. In your jotter, draw the circuit and using the correct terminology, explain exactly how this circuit works.



7. Part of a production line involves a quality check. If goods are seen to be faulty then they are pushed off the conveyor by a single-acting cylinder. Two people are used to make sure that no faulty goods leave the factory. They operate the cylinder by pressing a button.



(a) Design a circuit to solve this problem.
(b) Build and test your solution.
(c) In your jotter, draw the circuit and using the correct terminology, explain exactly how this circuit works.
(d) Draw a truth table of your results.
(e) Someone has suggested changing the shuttle valve to a T-piece. Why is this not a good idea? (You might want to build this circuit to investigate.)

8. A bus door is operated by pneumatics. The door is operated by a single-acting cylinder and controlled by a 3/2 valve. In an emergency, there should be a second valve that allows passengers to open the doors.



(a) Design a circuit to solve this problem.

(b) Build and test your solution.

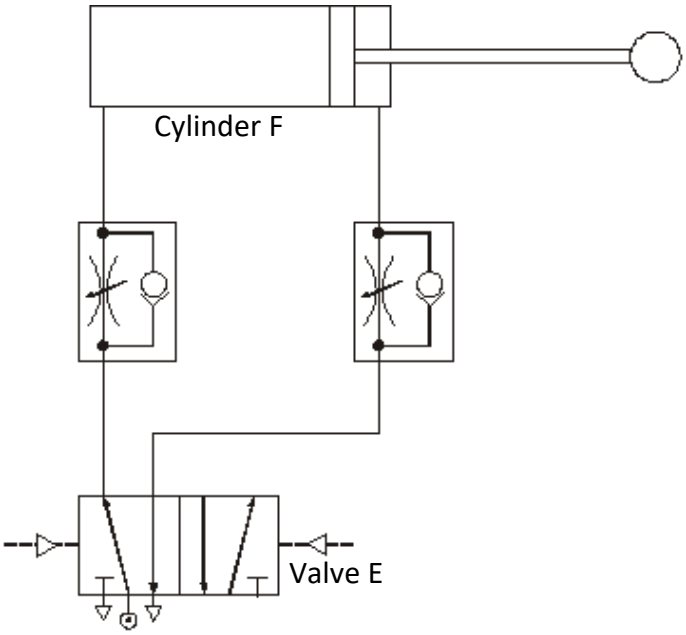
(c) Explain your choice of actuators.

9. A pneumatic circuit has been devised for use in operating a sliding door. It must be possible for the door to be opened or closed from both the inside and outside. The speed of the door should be controlled when opening and closing. A diagram of the components is shown below with some of the piping missing.

a) Complete the diagram.

b) Build and test the system.

c) Name all of the components used.



Valve A:

Valve B:

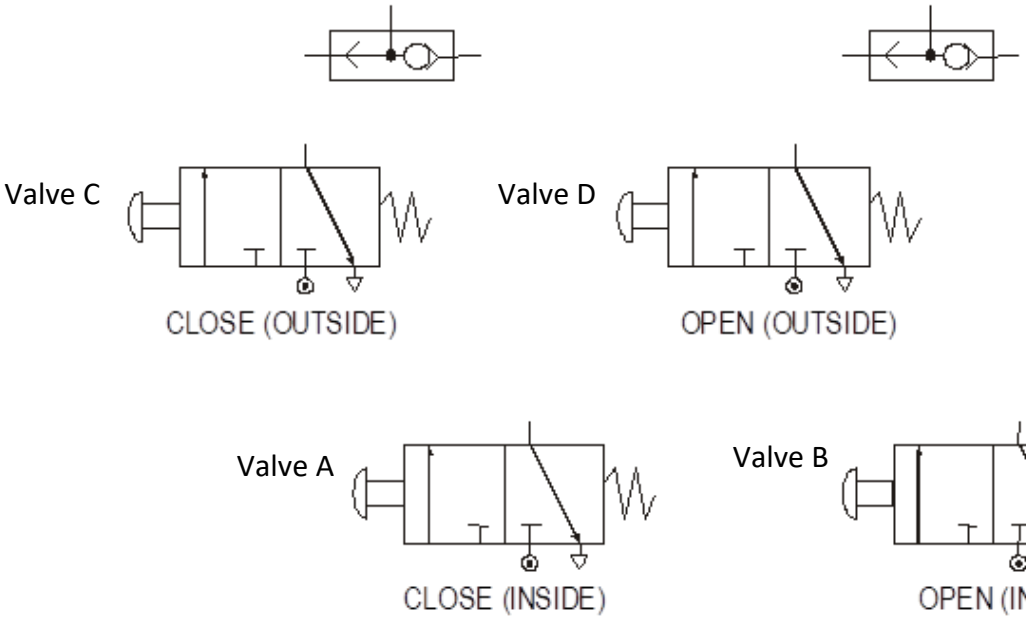
Valve C:

Valve D:

Valve E:

Cylinder F:

d) What type of control does this circuit use?



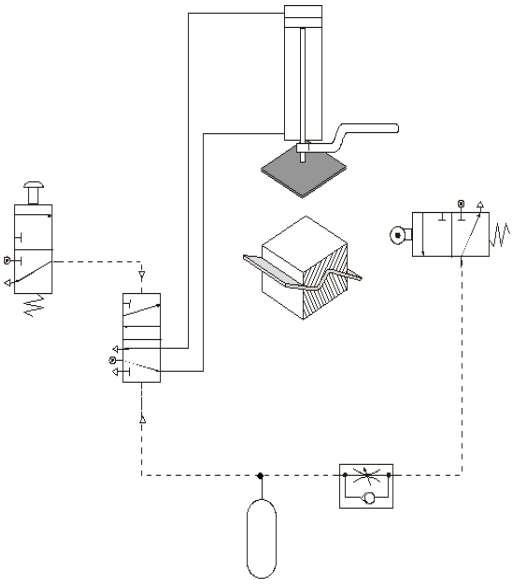
Automatic circuits

Automatic circuits are commonly found on production lines. They help to speed up production and make sure that the goods are all manufactured to the same standard.

There are two types of automatic circuit: semi-automatic and fully automatic.

Semi-automatic

A semi-automatic system only completes the process once. It requires a physical action for it to start.



In the system shown, it only starts by the operator actuating the push button, 3/2 value, spring return valve. This sends a pilot air signal to the 5/2 valve and it changes state. This allows new mains air to enter the system and causes the cylinder to outstroke. Once fully outstroked, it actuates the roller valve which sends a pilot air signal into the unidirectional restrictor. This device slows the speed of the air going into the reservoir which also causes a time delay (holds the piston rod in place while the material mould is taking shape. When the reserve is full, pilot air then continues into the other 5/2 valve to change its state and allows new mains air to enter the system and causes the cylinder to instroke. The system resets and stops.

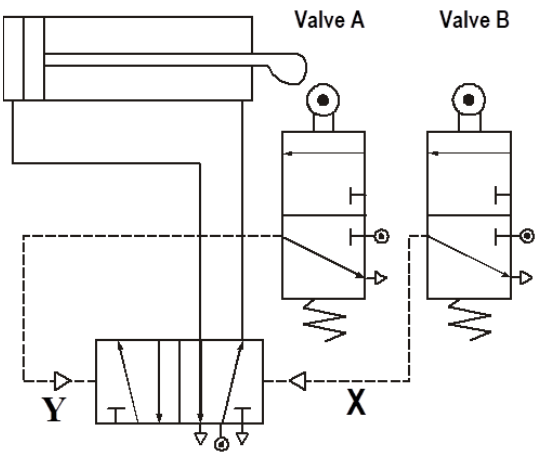
Fully automatic

A fully automatic circuit is one that continues to work, performing a task over and over again. It does not stop or wait for input from an operator. These circuits make use of actuators such as a roller trip and plunger to detect the position of the piston as it instrokes and outstrokes.

Automatic circuits produce reciprocating motion. This is motion up and down like the needle on a sewing machine. It can also be left and right, or forwards and backwards along a straight line. We can represent reciprocating motion by arrows like these: For example, a polishing machine requires the reciprocating motion of a double-acting cylinder.

Here is a circuit example and explanation of a fully automatic circuit; (3 marks)

"As the piston instrokes, it actuates valve A and allows pilot air (Y) into the 5/2 valve which changes state. This allows new mains air to enter the system and causes the cylinder piston to outstroke (1). When fully outstroked, it actuates valve B and causes pilot air (X) to change the state of the 5/2 valve. New mains air enters the system and causes the cylinder piston to instroke (1). The system is fully automatic and therefore the process begins all over again and continues to operate without stopping" (1).

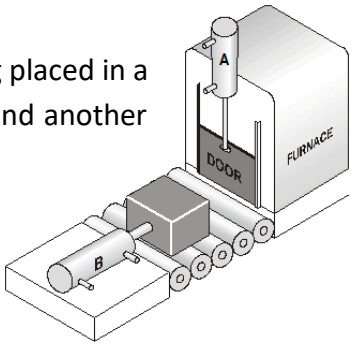


Sequential control

Many pneumatic systems and machines are designed to perform a range of tasks in a certain order or sequence. This usually involves the use of two or more cylinders working together to complete the task.

Example

A company has automated its production line that involves metal blocks being placed in a furnace for heat treatment. One cylinder is used to open the furnace door and another pushes the metal block into the furnace.



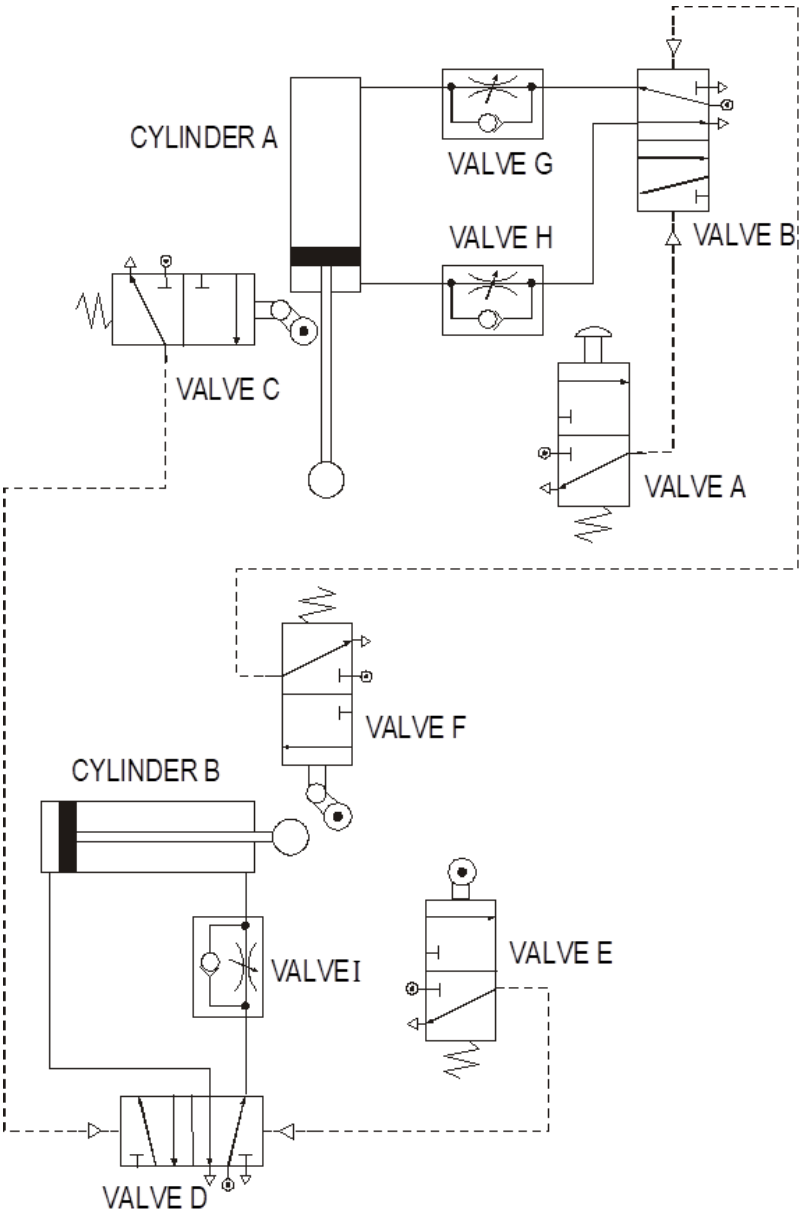
The sequence of operations for this process is as follows.

a) An operator pushes a button to start the process.
b) The furnace door is opened
c) The block is pushed into the furnace and the piston in strokes
d) The furnace door is closed
e) The sequence stop.

Below is the explanation and circuit diagram for the system; (6 marks)

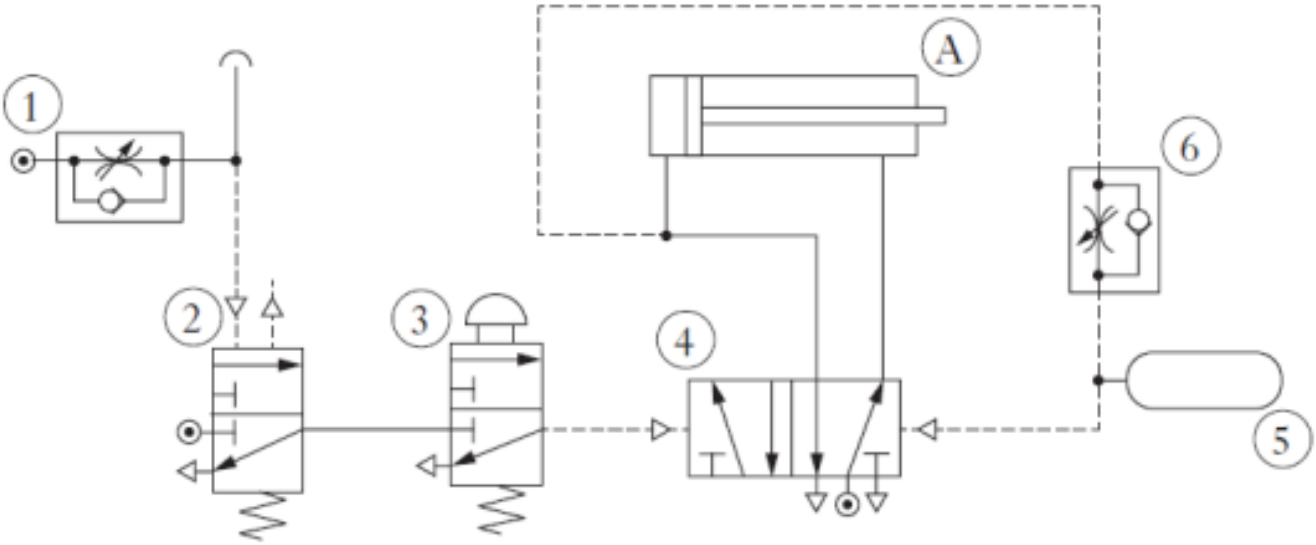
The system begins by actuating valve A. This changes the state of the valve B and new main air enters the system. This causes cylinder A to instroke slowly (valve G) (1), raising the furnace door. When cylinder A is fully instroked (negative), the piston actuates valve C and this send a pilot air signal to valve D (1)."

The 5/2 valve changes state and new main air enters the system and causes cylinder B to outstroke (positive) slowly (valve I) (1). The block is pushed into the furnace. When cylinder B is fully outstroked, the piston actuates valve E and pilot air changes the state of Valve D. New mains air enters they system to instroke the cylinder B (1). When Valve F is actuated by cylinder B instroke, a pilot air signal is sent to valve B to change the 5/2 state which causes cylinder A to outstroke slowly (valve H) (1). Cylinder A stays in the positive position which keeps the furnace door closed. The system resets and stops. It will only start again when valve A is manually actuated (1).



10. A pneumatic circuit used in a production line as shown below. Build and test the circuit below.

(*a*) State the **full** name of the following pneumatic components.



Component 1:

Component 2:

Component 3:

Component 4:

Component 5:

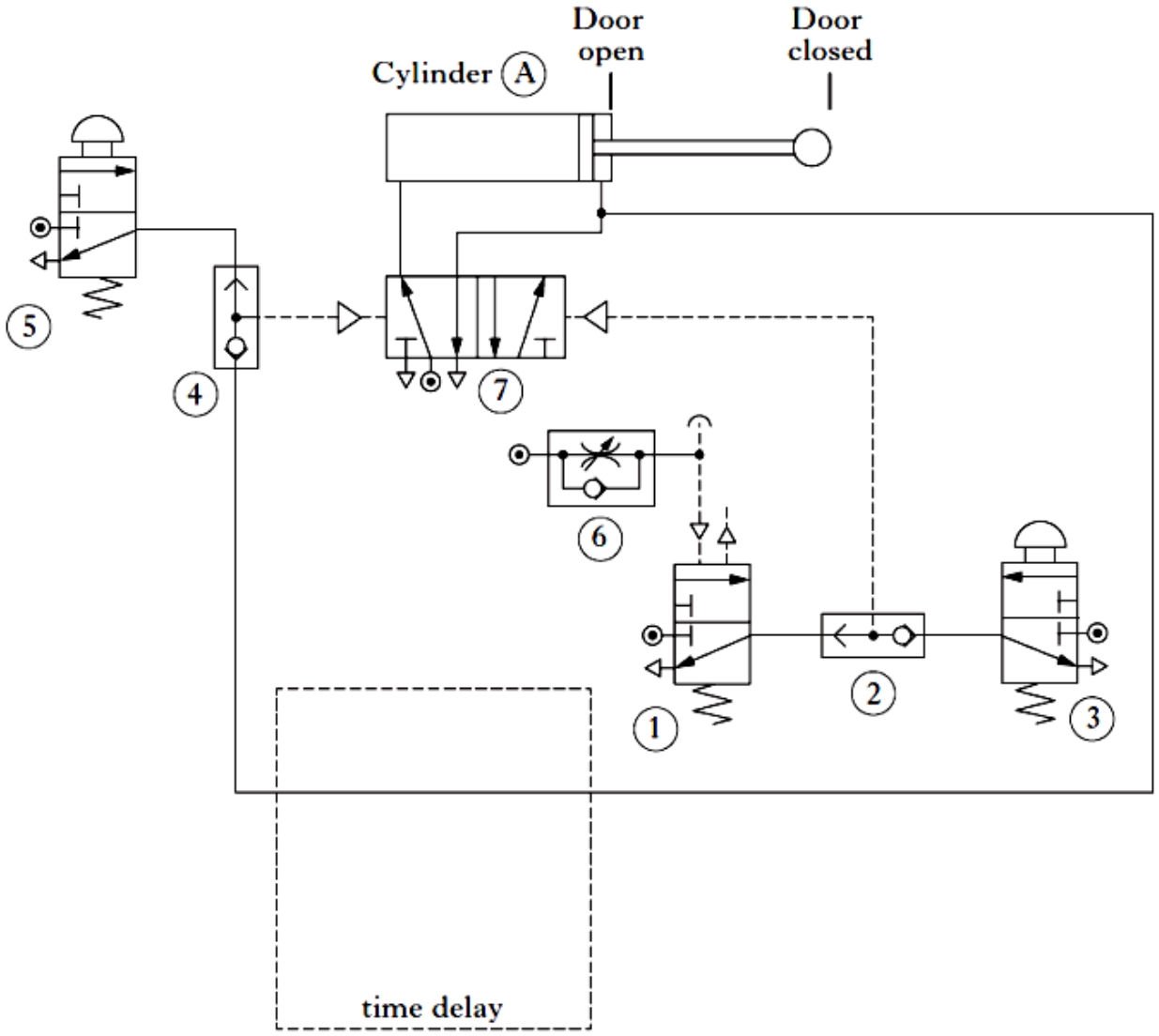
Component 6:

Component A:

(*b*) Describe, using appropriate terminology, the operation of the pneumatic circuit.

An increase in pressure is sensed …

11. The pneumatic circuit for a door system is shown below.

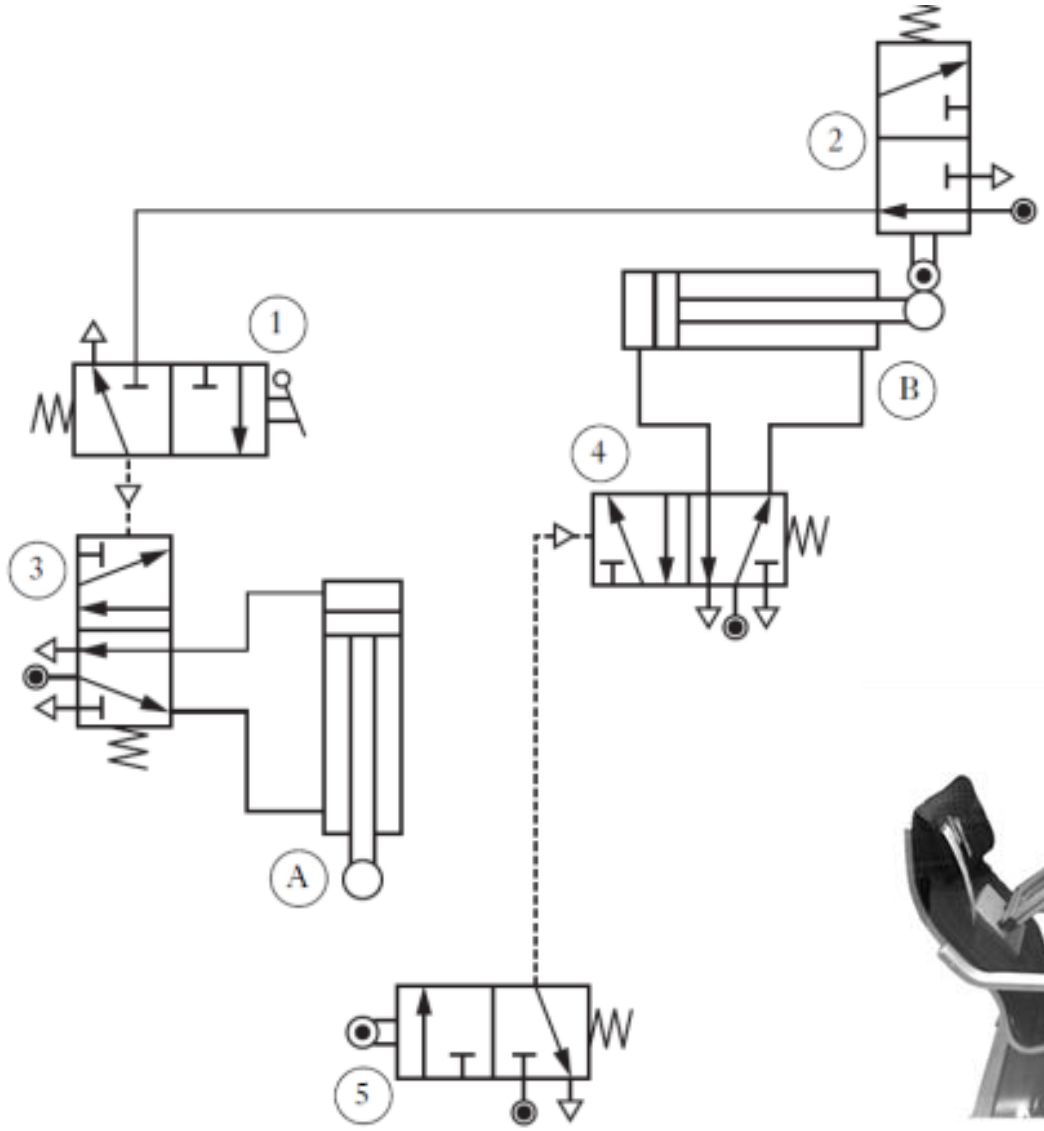


(*a*) State the name of component 1:

(*b*) Draw, in the position shown in the above circuit the components required to create a controlled time delay when closing the door.

(*c*) Describe, using appropriate terminology, the operation of the door system.

12. The pneumatic circuit shown is used to test the life span of a chair.





(*a*) State the **full** Name of the following pneumatic components.

Valve 1:

Valve 3:

(*b*) Describe the operation of the pneumatic circuit for the test chair.
When Valve 1 is actuated….

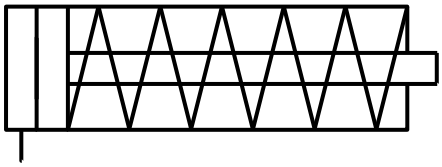
Forces in a single acting cylinder

When a single-acting cylinder outstrokes, it exerts a force.

We can use this force to carry out tasks. When designing pneumatic circuits, we must use a cylinder that is capable of completing its task. For example, if a single-acting cylinder is used to push parcels off a conveyor belt, then it must produce a big enough force to be able to do this. If the force is not big enough, the parcels will not move, and if the force is too big, the parcels may be damaged.

The size of the force produced by the cylinder as it outstrokes depends on two things- air pressure applied to the cylinder and the surface area of the piston. This means that if we want a bigger force *we can either use a larger piston or increase the air pressure.* However, it is not a good idea to increase the air pressure because this can cause damage.

The instroke of a single-acting cylinder is controlled by a spring. The spring returns the piston to its original position. We do **not** normally use the instroke of a single-acting cylinder to carry out tasks.

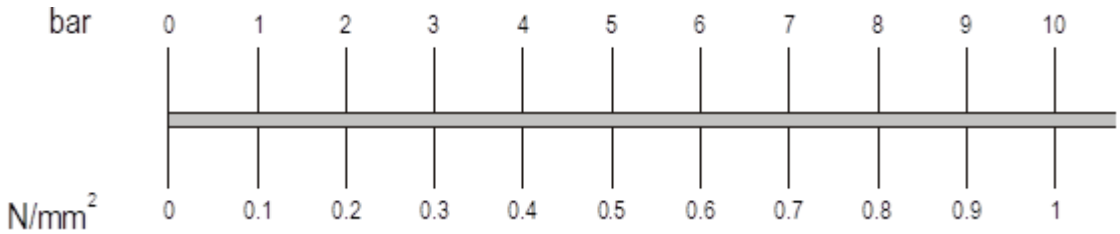


Pressure

Air pressure is measured in bars or in N/mm^2 (newtons per square millimetre). We can measure the pressure in a pneumatic system using a pressure *gauge*. A gauge will always be connected to the compressor, but other gauges may be connected throughout large systems. This helps to detect leaks, as the pressure in the system would begin to fall if air was escaping from the pipes.

Whenever we use pressure in calculations, we require the units to be in N/mm^2. This sometimes means converting from bars to N/mm^2. This conversion is easy, as you simply divide the value in bars by 10. For example, if the pressure supplied to a system is 5 bars, we can find the equivalent value in N/mm^2 by simply dividing 5 by 10. Therefore, the value would be 0.5 N/mm^2 (2 s.f.)

The chart below provides a quick reference.



Area

The surface of the piston is the area that the air pushes against to outstroke the piston. This area is circular. The area of a circle is calculated using the formula;

$$\textbf{Area} = \pi r^2 \text{ or } \frac{\pi d^2}{4}$$

'r' is the radius and ***'d'*** is the diameter of the circle.

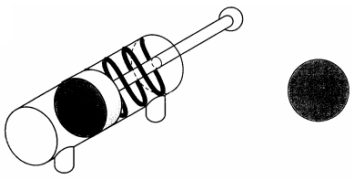
Force

The force produced when a **single-acting cylinder** outstrokes is calculated using the formula;

Force = Pressure x Area

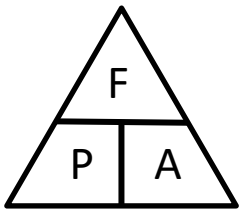
- **Where Force is measured in Newtons**
- **Pressure in N/mm²**
- **Area is measured in mm².**

In some situations, we would know the size of the force needed to do a job properly. In this case, we would want to calculate the pressure needed or the size of the piston. To do this we need to rearrange our formula.



$$Pressure = \frac{Force}{Area}$$

$$Area = \frac{Force}{Pressure}$$



Worked example for single acting

Air is supplied to a single-acting cylinder at a pressure of 3.0 N/mm² (2 s.f.) The diameter of the piston is 25 mm. Calculate the force produced as the piston outstrokes.

Step 1

Write down any information that you have from the question.

Pressure = 3 N/mm²

Diameter = 25 mm

Step 2

We need to calculate the surface area:

$$Area = \pi \frac{d^2}{4}$$

$$= 3.14 \times \frac{25^2}{4}$$

Step 3

$$= 491 \text{ mm}^2$$

Use the correct formula for what you are trying to find. In this case, calculate the force.

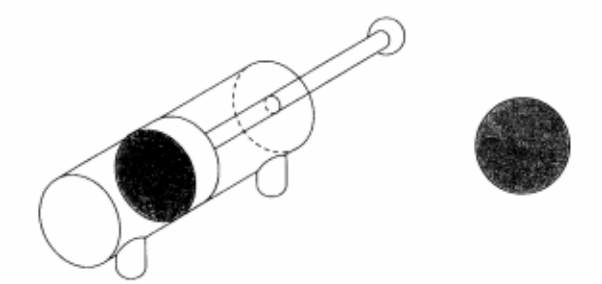
Force = Pressure × Area

Force = 3.0 × 491 = 1473 N (4 s.f.) Final answer of **Force = 1.47 kN (1.5KN for 2 s.f)**

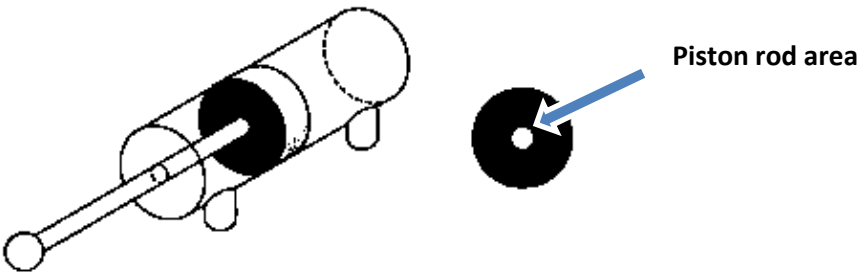
Forces in a double-acting cylinder

We already know a double-acting cylinder can be more useful to us because the input and output strokes are controlled by compressed air. This allows us to make use of both the outstrokes and instrokes force.

What we learn however is that the outstrokes force is **greater** than the instroke force. During the outstroke, the compressed air pushes against the surface area of the piston in the same way as in the single-acting cylinder.



However, during the instroke the surface area is **reduced** because of the piston rod. This means that the compressed air does **not have as big an area to push against and so it does not produce as big a force**.



We can find this surface area, or ***effective area*** as it is known, by calculating the area of the piston rod and subtracting it from the surface area of the piston.

Effective Area = piston area – piston rod area

Worked example double acting

A double-acting cylinder has a piston with a diameter of 25mm. The piston rod is 5mm in diameter. Pressure is supplied to the system at 4.0 N/mm^2. Calculate the force produced by the cylinder as it outstrokes and instrokes.

Step 1

Write down any information that you have from the question.

Pressure = 4.0 N/mm^2

Piston diameter = 25mm

Piston rod diameter = 5mm

Step 2

We need to calculate the surface area:

$$Area = \pi \frac{d^2}{4}$$

$$= 3.14 \times \frac{25^2}{4}$$

$$= 491\,\text{mm}^2$$

Step 3

Use the correct formula for what you are trying to find. In this case, calculate the outstroke force.

Force = Pressure × Area

= 4.0 × 491

= 1964 N

Outstroke force = 1.96 kN

Step 4

Calculate the piston rod area:

$$Piston\ rod\ area = \pi \frac{d^2}{4}$$

$$= 3.14 \times \frac{5^2}{4}$$

$$= 20\ \text{mm}^2$$

Step 5

Calculate the effective area. (We already know the piston area from step 2)

Effective area = Piston area – piston rod area

= 491 – 20

= 471 mm^2

Step 6

Calculate the instroke force.

Force = Pressure × Effective Area

= 4.0 × 471

= 1884 N (Do not leave your answer as this with 4 significant figures as it WILL loose the final mark)

Instroke force = 1.88 kN (1.9KN 2.s.f.)

Calculation questions 1

1. Write down the formula that we use to calculate the force in a single-acting cylinder as it outstrokes.

2. A pneumatic stamping machine is used to stamp the company logo onto metal casings. It is discovered that the stamp does not imprint the logo properly. Suggest ways of increasing the size of the force produced by the cylinder.

3. What controls the instroke of a single-acting cylinder?

4. A single-acting cylinder is used to press two sheets of acrylic together when they are gluing. The process requires a force of 300 N. The only piston available has a diameter of 20 mm and it is supplied with air at a pressure of 0.3 N/mm^2. Will this arrangement enable this process to be carried out properly?

5. What force will be produced by a 20 mm diameter cylinder as it goes positive using a pressure of 0.8 N/mm^2?

6. Calculate the outstroke force produced by a 40 mm diameter cylinder when it operates with a supply pressure of 3 bars.

7. Write down the formula that we would use to calculate the pressure of a system if we already knew the force required and the size of the cylinder available.

8. A stamping machine exerts a force of 454 N with a piston diameter of 34 mm. Calculate the air pressure required for this operation.

9. A machine that places tops on bottles uses a single-acting cylinder. The process requires a force of 650 N. What air pressure needs to be supplied to the cylinder with a diameter of 56 mm?

10. A force of 540 N is needed to push a packing case off a conveyor belt. The single-acting cylinder used has a diameter of 60 mm. What air pressure should be supplied to the system?

11. A pneumatic system is used to test the quality of drawer guides in kitchen cabinets. A force of 16 N is needed to open the drawer. The single-acting cylinder available has a piston diameter of 10 mm. What air pressure should be supplied?

12. Write down the formula we would use to find the area of a piston if we already knew the size of the force it needed to produce and the air pressure being supplied.

13. A single-acting cylinder is used to lift parcels on to a conveyor. This requires a force of 180 N with the system operating at a pressure of 6 bars. Calculate the area of the piston required.

14. A door requires a force of 400 N to slide it open. A single-acting cylinder supplied with a pressure of 5 bars controls the operation. Calculate the diameter of the piston required to produce this force.

    Use the formula: $Diameter = \sqrt{\frac{4 \times Area}{\pi}}$

15. A furnace door weighs 100 N and is lifted by a single-acting cylinder as it outstrokes. Compressed air is supplied at a pressure of 4 bars. Calculate the diameter of the piston required to raise the door.

    Use the formula: $Diameter = \sqrt{\frac{4 \times Area}{\pi}}$

Past paper questions

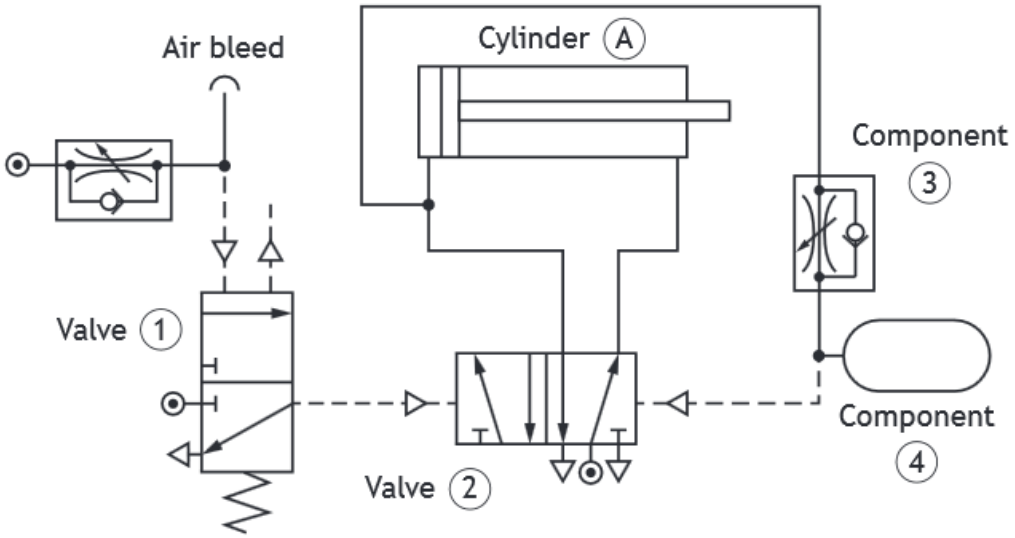
Review your Success Criteria at the front of the theory booklet and use the following tasks (A-K) to make any identified areas of developing knowledge to become secure knowledge. You can complete the tasks in any order.

Task A

A pneumatic circuit used in a production line as shown below.

Portions of the pastry travel along a conveyor belt where a pneumatic piston presses them into pie casings.

The pneumatic circuit shown below operates the piston when the pastry is sensed in position.



(*a*) State the **full** name of the following pneumatic components.

Component 3:

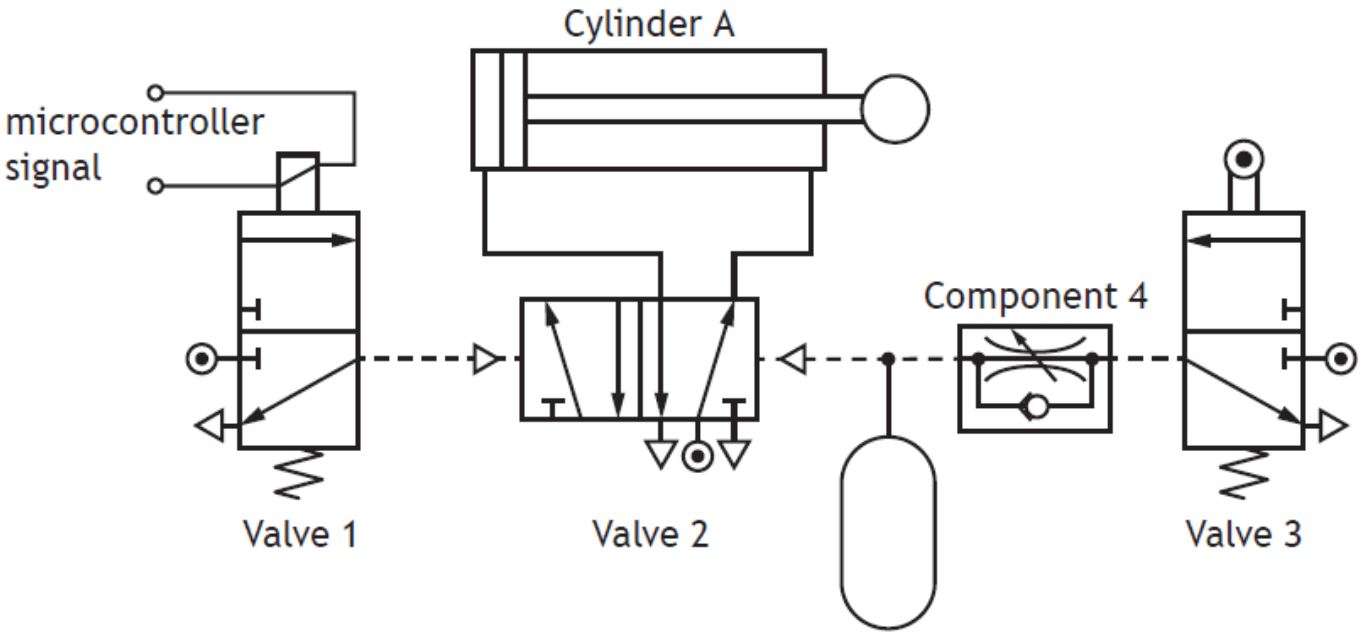
Component 4:

(*b*) Describe, using appropriate terminology, the operation of the pneumatic circuit.

When the air bleed is covered, value 1 is actuated . . .

Task B

A mechanical model of a dinosaur uses pneumatics to control its movement. Part of the pneumatic circuit used in the model is shown below.



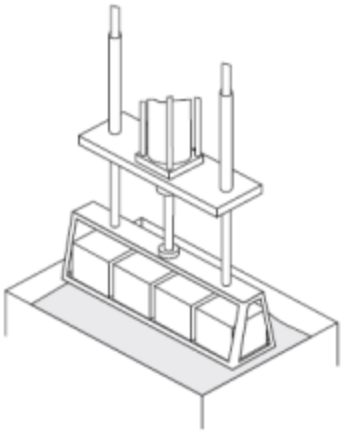
a) Describe, using the appropriate terminology, the operation of the circuit.

When a signal is sent from the microcontroller…

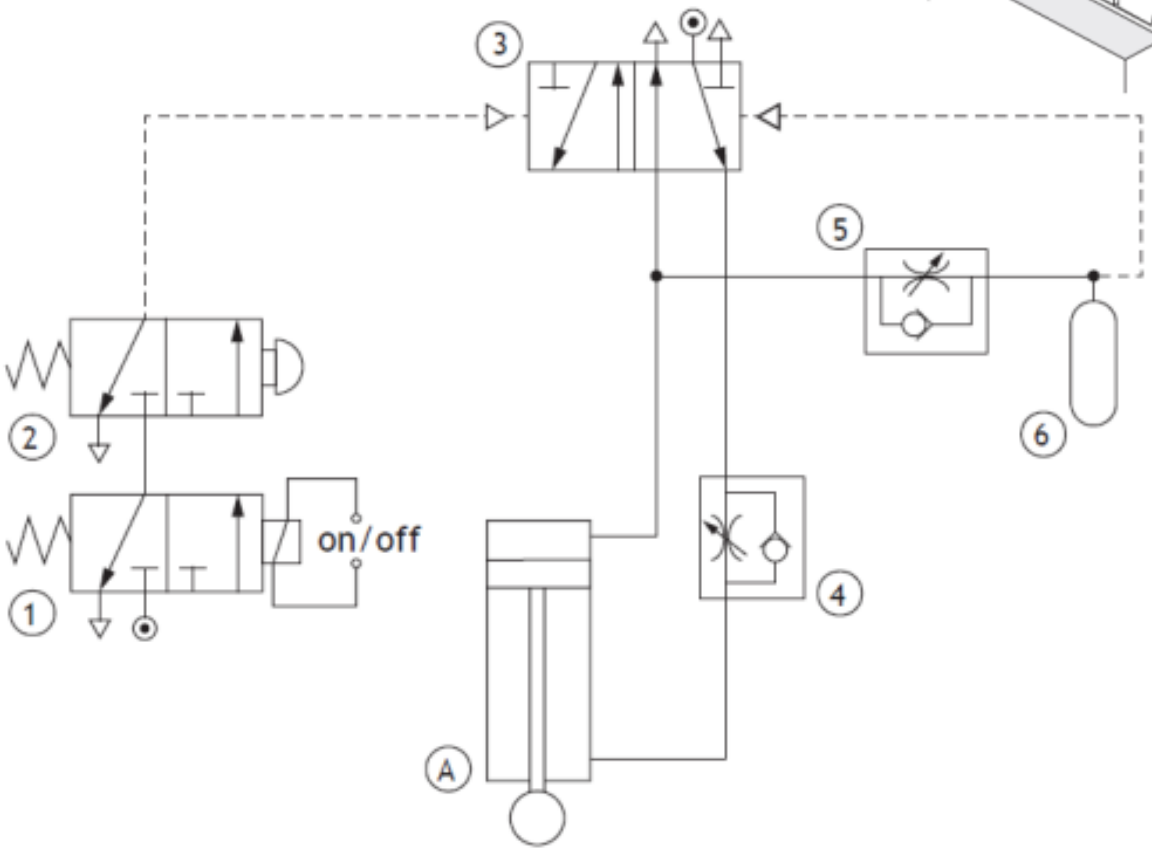
b) The circuit is to be altered so that the piston instrokes slowly. Indicate, with an **X**, on the pneumatic circuit where a unidirectional restrictor should be connected.

Task C

A pneumatic system used to lower metal components into an acid bath to be cleaned is shown below.

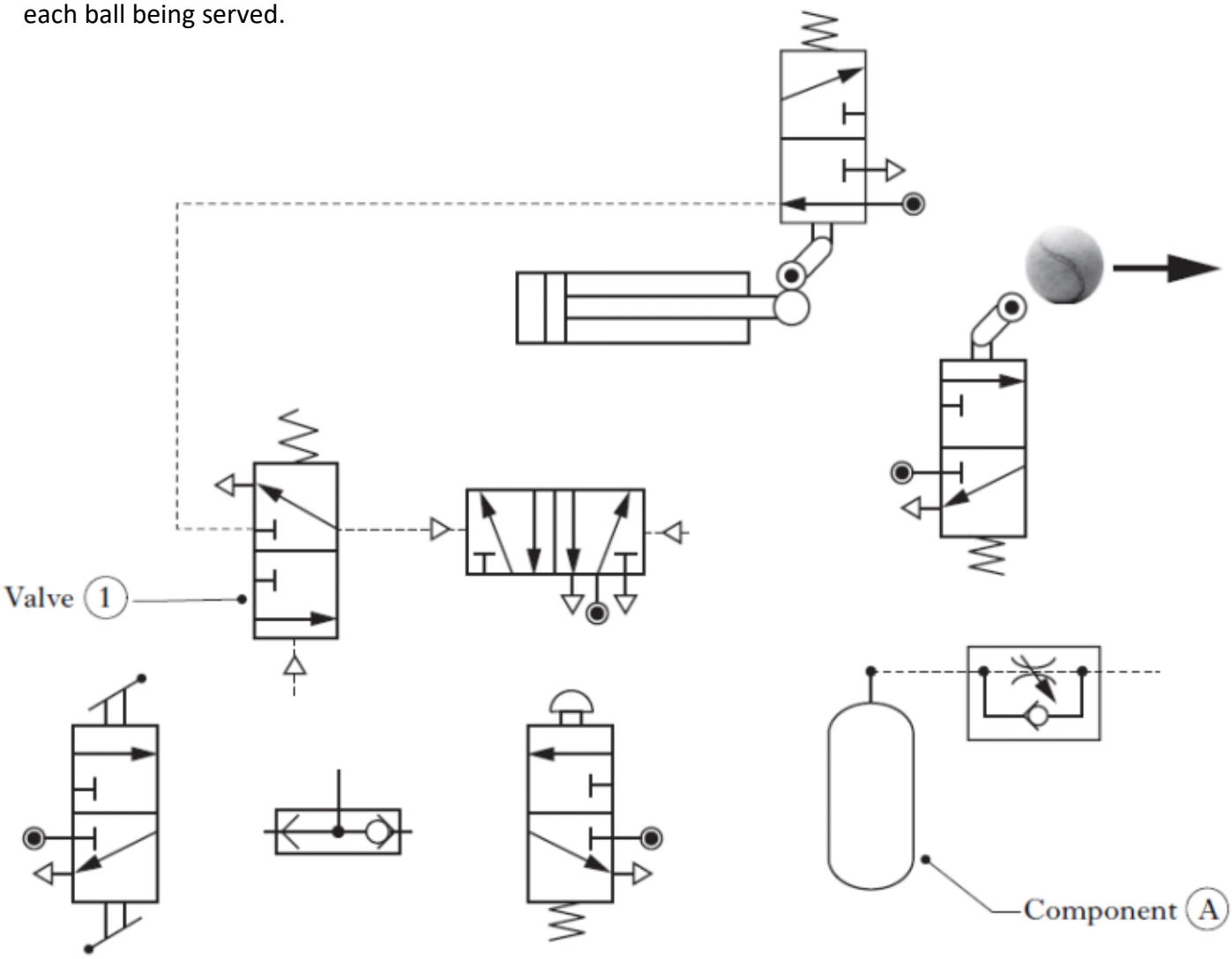


a) Describe the operation of the pneumatic circuit.



Task D

A pneumatic circuit is used to serve tennis balls during practice sessions. The system will serve automatically when a lever is actuated or manually each time a button is pressed. There is a delay between each ball being served.



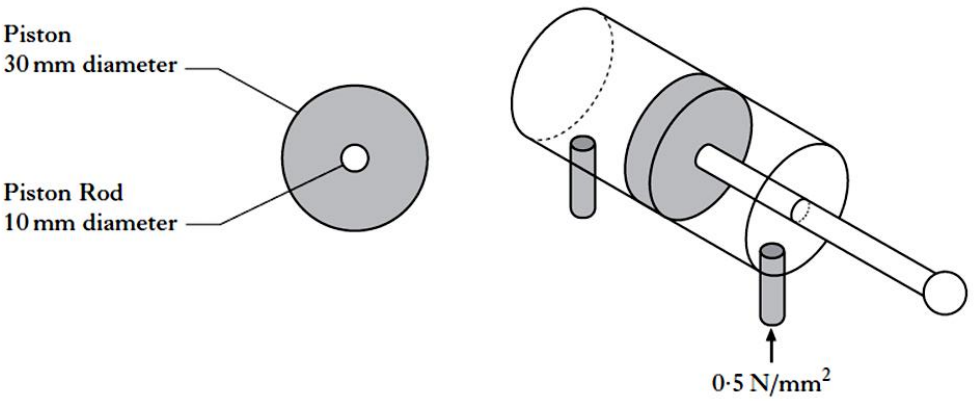
(*a*) Complete the piping of the pneumatic circuit.

(*b*) State the full name of the following components.

Component A:

Valve 1:

The piston below **instrokes** when air is supplied at a pressure of 0.5N/mm^2 to the cylinder.



(c) Calculate the in-stroking force.

(*d*) Describe **two** ways of reducing the **out-stroking** force applied by a piston.

1) _____

_____

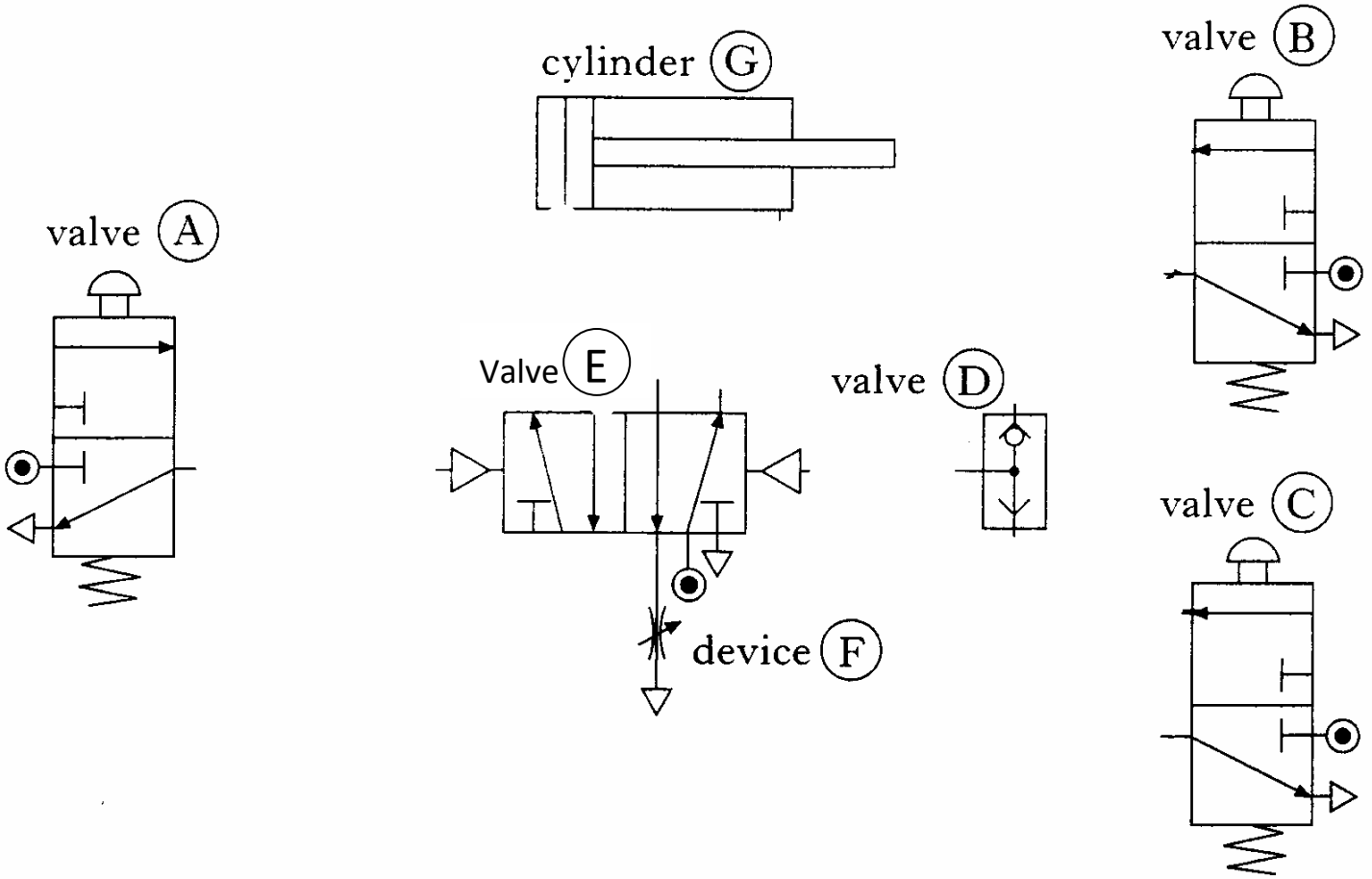
2) _____

_____

Task E

When valve A is operated, the piston out strokes and the platform is raised. When valve B or C is actuated, the platform will lower slowly.

a). Complete the piping of the pneumatic circuit below.



b). Give the **full name** of the following pneumatic components.

Value D:

Valve E:

Valve F:

Cylinder G:

c). Tick the correct answer. Valve D allows

NOT control ☐

AND control ☐

OR control ☐

Speed control ☐

Task E - Continued

d). Suggest a reason why there are 2 valves to lower the platform.

_____

_____

_____

e). Device F is piped into the circuit to slow the exhaust air leaving the cylinder when the piston **in strokes**.

Name another pneumatic device that could be used to **slow down the speed** of the piston in **one direction**.

---

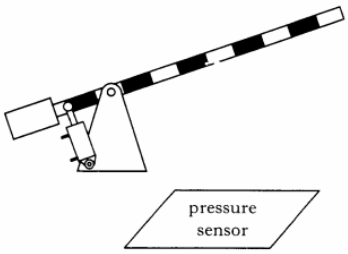
f). Sketch the symbol for this device from question 5.

g). Indicate with an '**X'** on the circuit in question 1, where this device should be placed.

h). Suggest a reason why the platform must be lowered slowly.

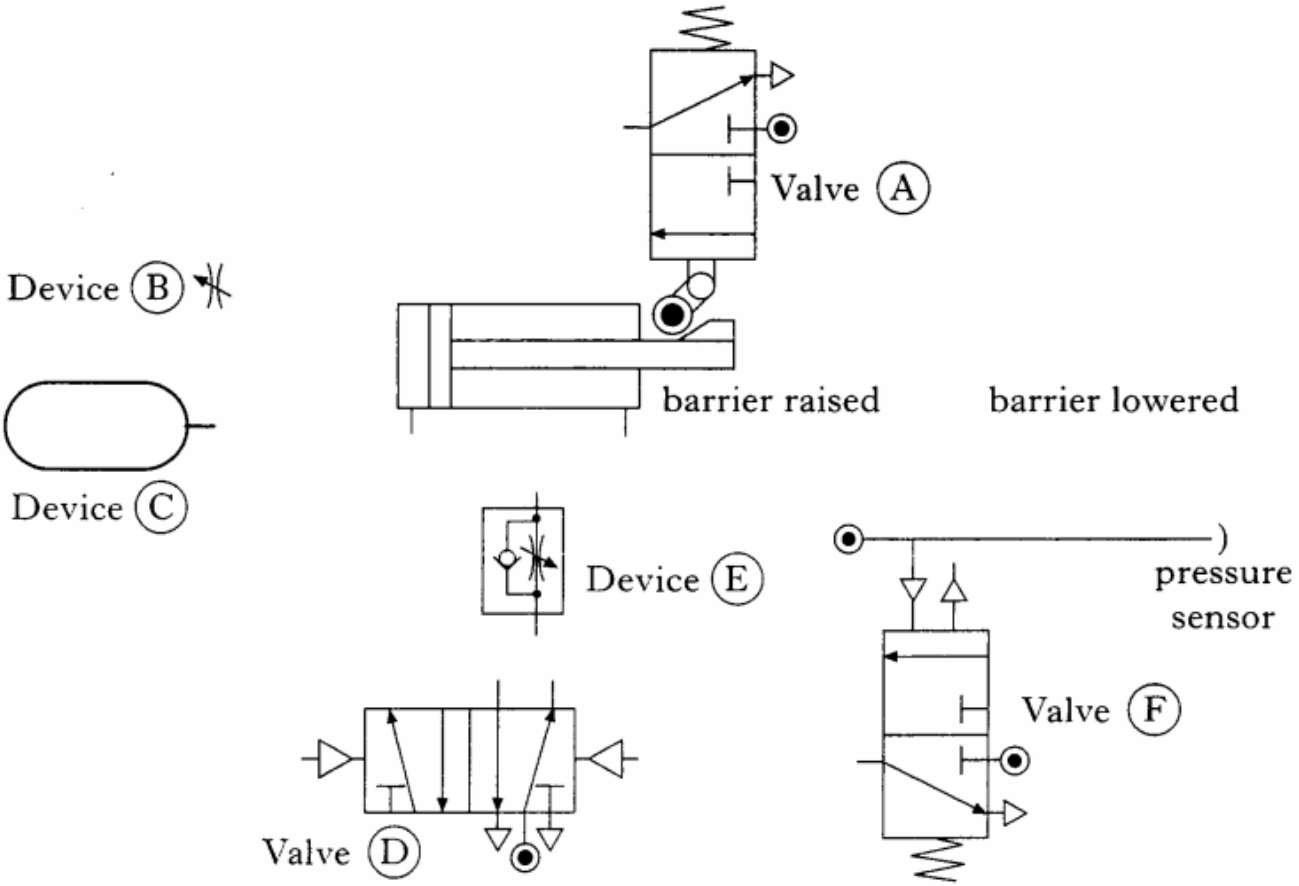
Task F

A pneumatic circuit is used to operate the entrance barrier of a car park.



If a vehicle drives over the pressure sensing area, the piston will **in stroke** and raise the barrier. After a 10 second **delay**, the piston will automatically **outstroke** and **slowly** lower the barrier.

a). Complete the piping of the pneumatic circuit.



Valve **F** actuator_____

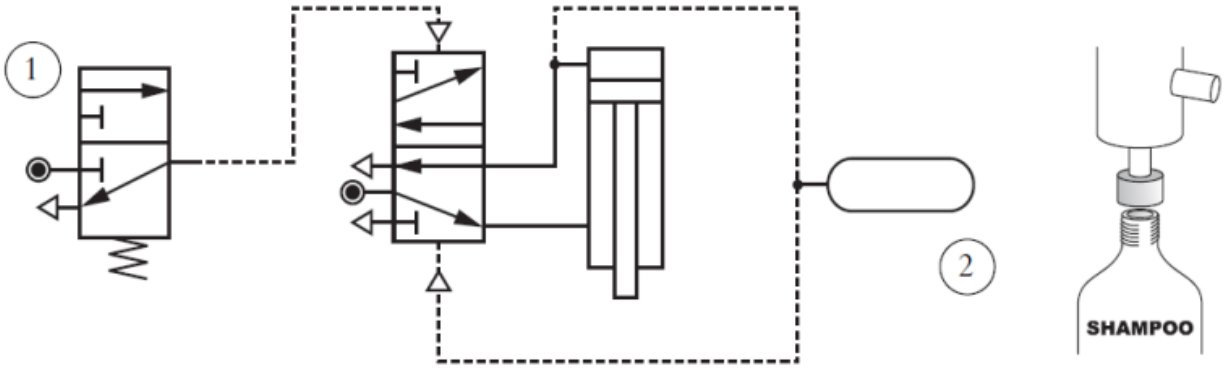
b). State the purpose of Devices **B** and **C** in the circuit.

---

c). Explain how the function of Device **B** differs from Device **E**.

Task G

Below shows an incomplete pneumatic circuit used to fix tops on shampoo bottles



Valve 1 is to be actuated by an electrical signal.

a) i) State the name of this actuator.

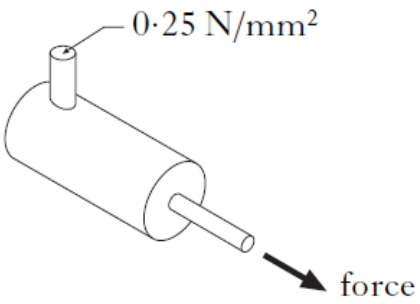
ii) Sketch the symbol for this actuator below.

b) i) Sketch on the circuit above the pneumatic device that will create an adjustable time delay after the piston outstrokes.

ii) State the name of component

A 10 mm diameter cylinder has air supplied at a pressure of 0.25N/mm^2.

c) Calculate the force of the piston as it outstrokes.

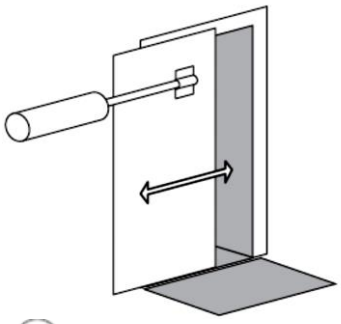


d) Explain why the in-stroke force of the piston will be **less** than the out-stroke force.

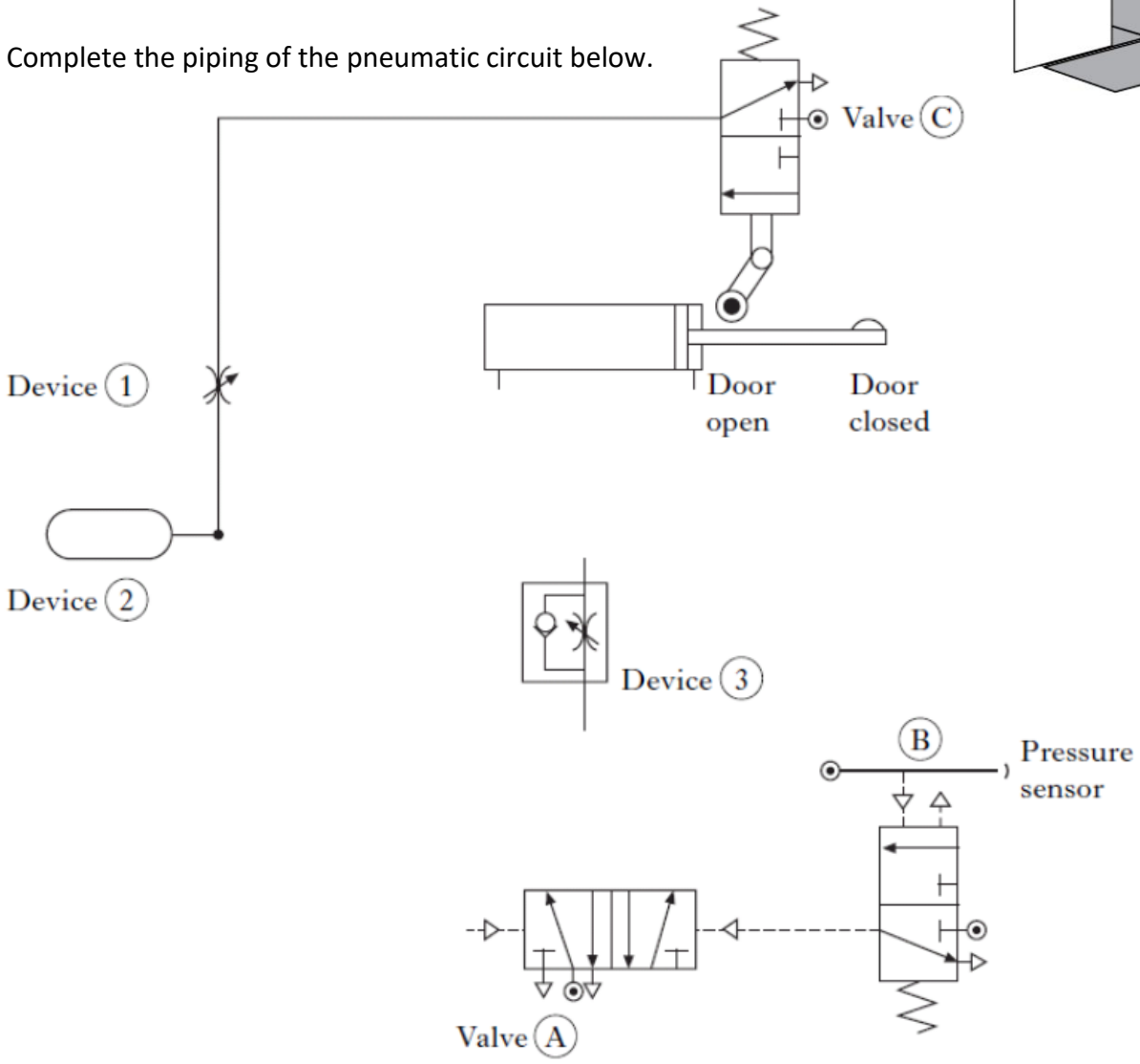
e) State a reason, other than cost, for using pneumatic systems in an industrial environment.

Task H

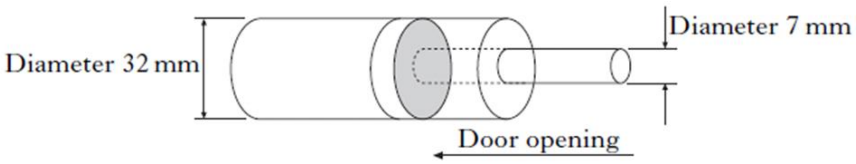
A pneumatic circuit is used to control the operation of an automatic door. When a person steps on to the pressure sensor, the piston will in-stroke and open the door. After an 8 second delay, the piston will automatically outstroke and slowly close the door.



a) Complete the piping of the pneumatic circuit below.

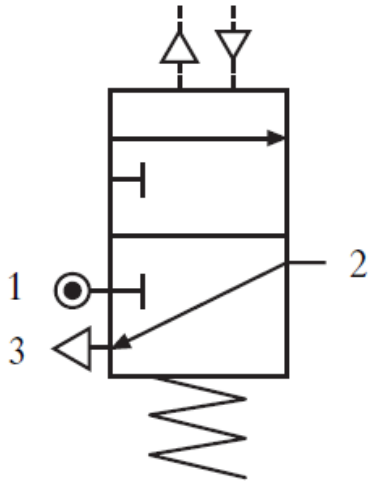


b) Calculate the in-stroking force when the air pressure is 2.32 N/mm^2.



Task I

The symbol for a pneumatic valve is shown opposite.

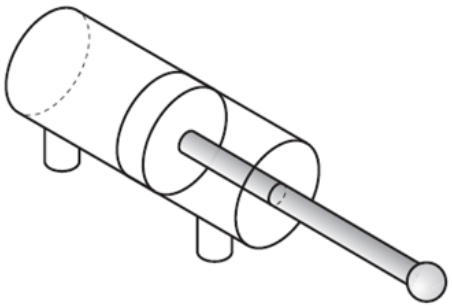


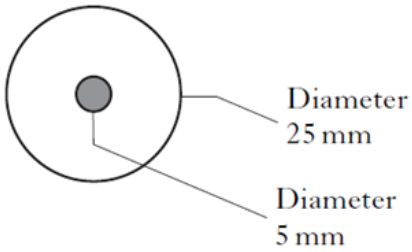
a) (i) State the **full** name of the pneumatic valve.

ii) Complete the table below for the valve ports.

Port	Connection
1	
2	Output port
3	

Below shows the dimensions of a double acting cylinder.





Calculate

b) (i) The effective area of the piston as it in-strokes;

ii) The **in-stroking force** of the piston if air is supplied at a pressure of 0·6N/mm^2.

TASK J

Complete the following task in your jotter.



A national zoo is setting up an enclosure for some new animals.
A system is required that will control a gate to allow the animals into the outdoor area. The gate should open when it is daylight or if a keeper activates a sensor. It should close when the keeper activates a separate sensor. Some parts of the system will be electronically controlled and other parts will operate using pneumatics.

a) Draw a pneumatic circuit that will perform the required functions. Assume that a 12 V signal will come from the electronic circuit when light level is high.

b) Construct your design and test it to ensure it performs the required functions.

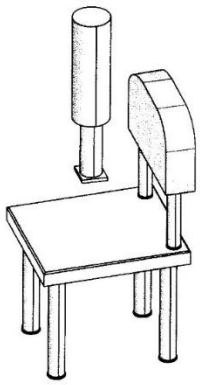
c) Describe how you tested the system and how it performed.

d) Compare the circuit's performance against the initial specification, and suggest how it could be adapted or changed to work better.

e) Many animals are at risk from the effects of climate change and global warming. Describe two ways that engineering solutions have tried to reduce these effects.

Task K

Complete the following task in your jotter.



An engineering lab tests the materials and construction used in a range of furniture to see if they conform to the European Union's safety standards. As part of this process, the lab runs a series of automatic pneumatic test rigs which have reciprocating piston rods simulating different wear and loadings on the furniture.

Design a pneumatic circuit to control the operation of a furniture test rig. The sequence for the pneumatic circuit is as follows:

- A 5/2 valve will operate a double acting cylinder and cause the piston rod to out-stroke. (The speed of this out-stroking movement must be adjustable.)
- An actuator will sense when the piston rod has fully out-stroked and the 3/2 valve will send a signal to cause the piston to in-stroke at full speed.
- The sequence will repeat until the air supply is switched off.

a) Apply your knowledge to design a possible solution to the problem brief.

b) Build a model of your solution.

c) Describe the tests you carried out and how the solution actually performed. Suggest possible improvements or further work.

d) Describe how you tested the system and how it performed.

e) Compare the circuit's performance against the initial specification, and suggest how it could be adapted or changed to work better.

f) Many animals are at risk from the effects of climate change and global warming. Describe two ways that engineering solutions have tried to reduce these effects.

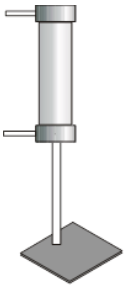
Calculations questions 2

1. Explain why the forces produced by a double-acting cylinder on the outstroke and instroke are different.

2. A double-acting cylinder found in a Technological Studies room has a piston diameter of 20 mm and is supplied with air at a pressure of 0.3 N/mm^2. What force is produced as the piston outstrokes? The piston rod has a diameter of 6 mm. What force is produced on the instroke?

3. A double-acting cylinder is used to raise and lower a barrier in a car park. The air pressure is 0.4 N/mm^2 and the piston has a diameter of 40 mm. The piston rod is 12 mm in diameter. What forces are produced when the piston outstrokes and instrokes?

4. A double-acting cylinder is used to set up skittles in a bowling complex. An instroking force of 0.04kN is needed to move the skittles. The effective area of the piston is 133 mm^2.



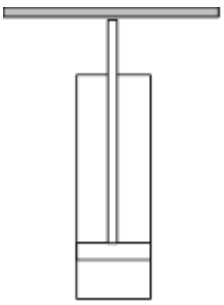
(a) What pressure should be supplied to the cylinder?

(b) At this pressure, will the outstroke force be larger or smaller than the instroke? Explain your answer.

5. Components on a conveyor system travel along and drop onto a table attached to the end of a double-acting cylinder. As the cylinder instrokes, the components are raised up and then pushed by another cylinder on to the next conveyor. The piston diameter is 20 mm and air is supplied at a pressure of 0.45 N/mm^2. The effective area is 200 mm^2.

(a) Calculate the instroke force and say whether the system could lift a component weighing 100 N.

(b) Someone suggests turning the double-acting cylinder round so that the components are lifted by the outstroke. Is the system now able to raise the components? Explain your answer.



Homework assignments

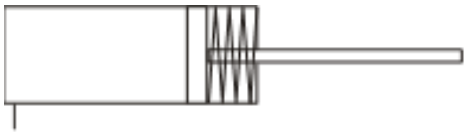
Homework 1

a) Describe some of the advantages of using pneumatics.

b) List the safety rules we must follow when operating pneumatic circuits.

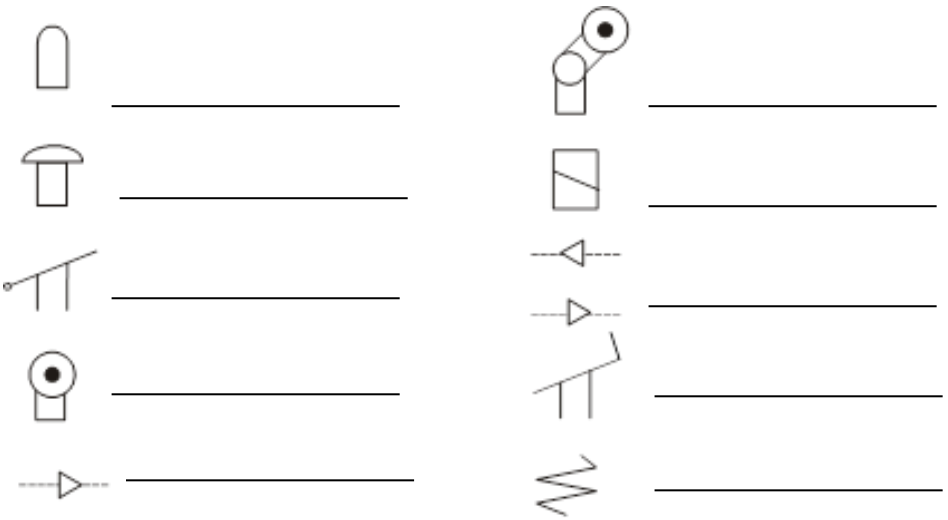
Homework 2

a) Which terms can be used to describe the piston in the position shown?



_____ _____

b) Name each of the following 10 actuators.



Homework 3

Draw the symbols for the following components.

a) Push button, spring return 3/2 valve

b) Single-acting cylinder

c) Plunger-operated, spring return 3/2 valve

d) Unidirectional restrictor

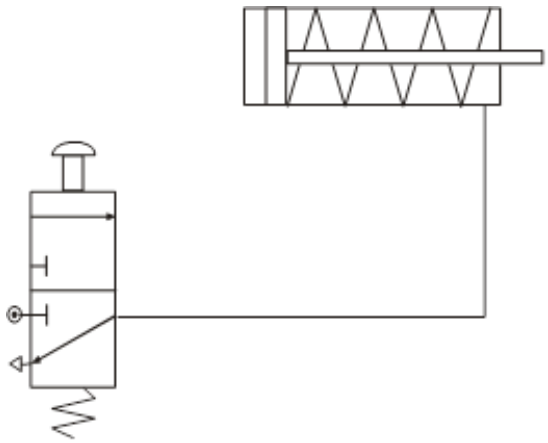
e) Double-acting cylinder

f) Draw the symbols used to represent main air and exhaust.

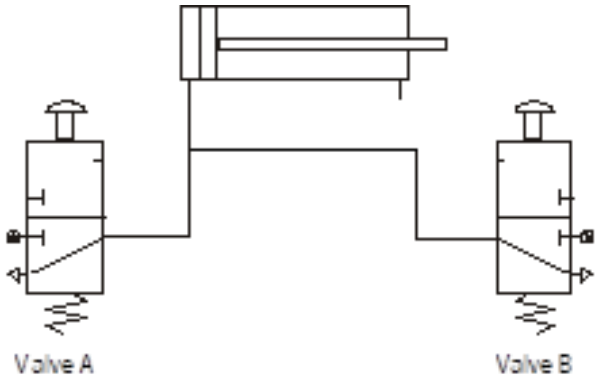
Homework 4

The diagrams below have a basic fault. Identify this fault and then redraw the diagram properly.

1)



2)



Homework 5

a) What is the difference between a restrictor and a unidirectional restrictor?

b) Why do we restrict the exhaust air from a cylinder rather than the air entering the cylinder?

c) Name the components used to create a time delay. Draw a diagram to show how they are connected together.

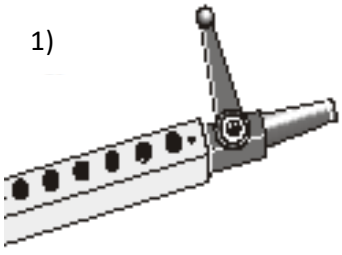
d) How can we change the length of a delay?

Homework 6

a) Describe the difference between a T-piece and a shuttle valve. You may use sketches to help.

b) Name the following components and describe what they are used for.

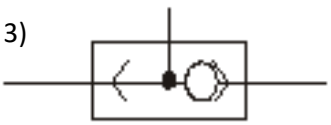
1)



2)



3)



Homework 7

a) A circuit allows a door to be opened by pressing either valve A or valve B. What type of control is this?

_____

b) Draw the truth table for this type of circuit.

c) How can AND control improve safety on a production line?

_____

_____

_____

_____

_____

d) Draw a pneumatic circuit that would demonstrate AND control.

Homework 8

a) What name is given to the movement produced by an automatic circuit?

_____

b) What is the difference between a semi-automatic and a fully automatic circuit?

c) Explain why the force produced by the instroke of a double-acting cylinder is less than the outstroke.

d) Safety barriers on a fairground ride are held in place by pneumatic cylinders. Which type of cylinder would you recommend? Describe the reasons for your choice.

Homework 9

a) A double-acting cylinder is used to open and close a window in a greenhouse. The window weighs 20 N and the piston diameter of the cylinder is 10 mm. What air pressure should be supplied to this system? (Show all your working.)

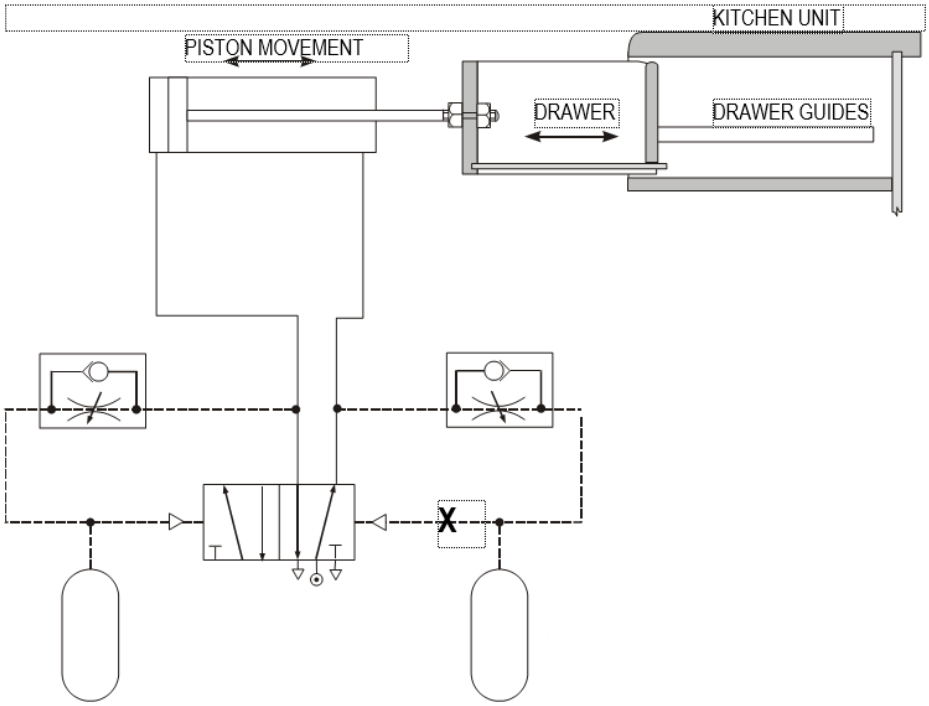
b) A force of 280 N is needed to tip over a container full of rubbish. Compressed air is supplied to the pneumatic system at a pressure of 0.7 N/mm^2. What cylinder diameter is needed to complete this task?

c) A pneumatic system is used to dip components into a chemical solution. It is no problem for the cylinder to lower the components, but it is unable to raise them by instroking. The components weigh 550 N and the cylinder has a diameter of 32 mm. The piston rod is 12 mm in diameter and air is supplied to the system at a pressure of 0.7 N/mm^2. Show by calculation why the system cannot lift the components.

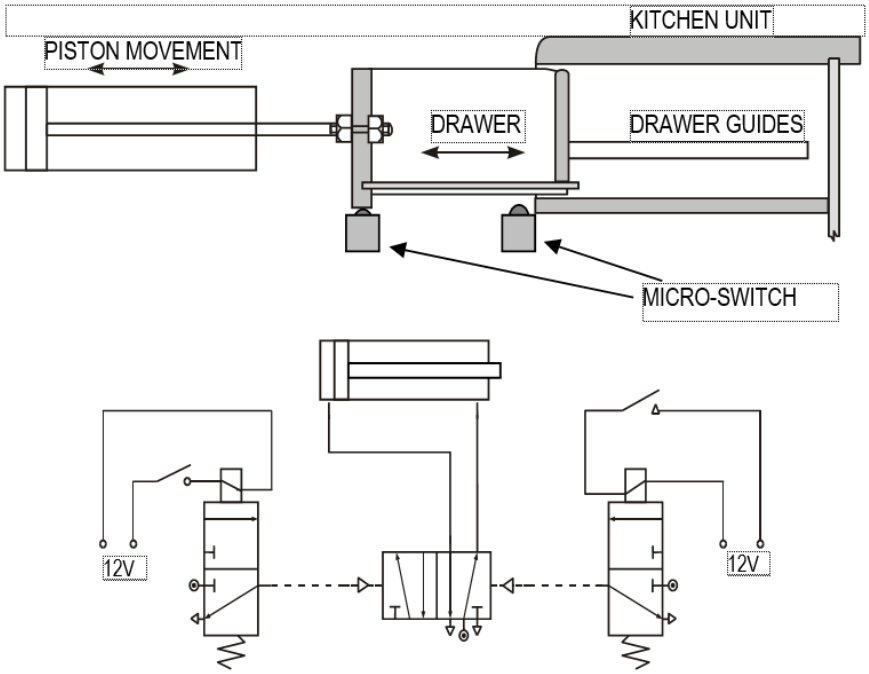
Homework 10

A furniture manufacturer uses pneumatics to test wear and tear on drawer guides in a kitchen unit. A double-acting cylinder is used to constantly open and close the drawer, and the runners are checked regularly. The following pneumatic circuits (A&B) have been designed to carry out the same task. Describe how each circuit works and choose the circuit that you think would best perform the task. Give reasons for your choice.

Circuit A



Circuit B



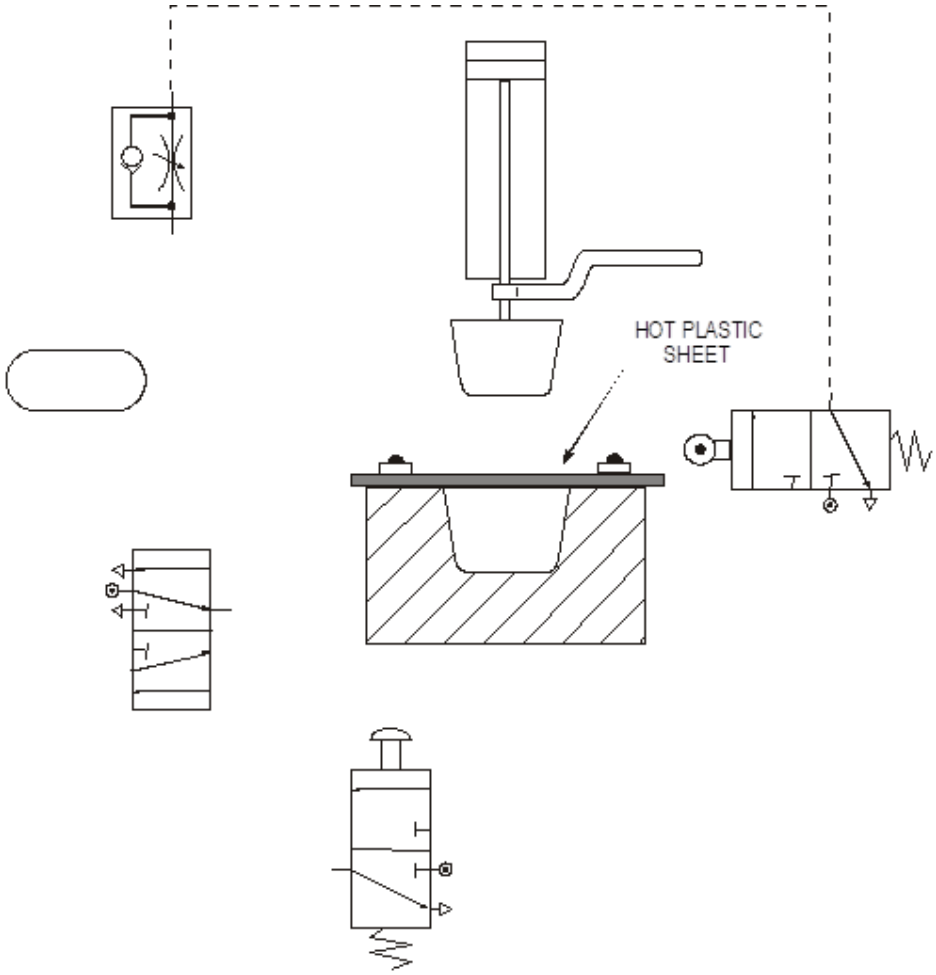
The best circuit is _____

_____

because _____

Homework 11

Hot plastic sheets are moulded into cups using a former attached to the end of a double-acting cylinder. An operator starts the process by pressing a button, but the cylinder should instroke automatically. Before this happens, there should be a delay to allow the plastic sheet to cool and set in shape. A partly completed diagram is shown.



a) Complete the missing piping.

b) State the full name of each of the pneumatic components marked on the diagram.

i) _____

ii) _____

iii) _____

c) Using appropriate terminology, explain how the circuit works.

_____

_____

_____

_____

d) The cylinder instrokes so quickly that sometimes the cup gets stuck onto the former. Name the component that could be used to reduce the speed of the cylinder.

e) The component named in your answer to (d) can be inserted in the circuit shown above so that the cylinder instrokes slowly. Add this component to the circuit diagram at the appropriate point.

f) Describe two ways of changing the length of time delay.

i)

ii)

g) It requires a force of 16 N to press the plastic sheet into shape. What air pressure is required if the diameter of the piston is 10 mm. (show all working and units.)
Pressure = Force/Area, Area = $\pi r^2 = \pi d^2/4$]